

On the Universality of Comparability Grids for Measurement-Based Quantum Computation

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Abstract

The measurement-based quantum computing (MBQC) model presents a way to do quantum computations in constant depth by measuring a pre-prepared graph state. As such, there is interest in determining which graph states are universal for MBQC. The universality of the comparability grid is currently an open problem, and of some interest as if it is universal, so are all families of graph states with unbounded rank-width. In this thesis, we report progress on this problem by discussing a variety of approaches for potentially proving and disproving the universality of the comparability grid for quantum computation.

Acknowledgements

First, I would like to thank Professor Daniel Grier for introducing me to the weird and wonderful world of quantum computing. His advice has been invaluable these past two years. He has been an incredibly kind mentor and I owe much of my growth, academically and personally, to him.

I would also like to extend this gratitude to Du (Dan) Van, the primary collaborator of this project, who has always been available to listen to my rambles. I wish him the best of luck in his future studies at NUS. I would also like to thank Jackson Morris, who has always been available to mentor and provide feedback for this project. I am also thankful for Kunal Marwaha, who laid some of the groundwork for this project, and kindly advised us during his visit to UCSD.

I also thank my family and friends for their utmost support in all aspects of my academic journey. It is through their efforts that I may focus on my studies and pursue my aspirations without distraction.

Preface to the Second Rough Draft

Incomplete sections and comments are written in blue throughout.

One of the great difficulties writing the rough draft was figuring out where to stop. In other words, how much can I assume from the reader? To answer this question, the past Mathematics honors theses of Ray Tsai, Clara Woods, and Jared Hughes in particular have been most helpful; the first for its recency and treatment of graph theory, the second for its experimental and quantum focus, and the third for its approach to 3XOR and theoretical computer science. These three, among other theses, appear to take a very introductory stance, writing for a non-expert audience familiar with only the typical mathematical fields (i.e. a senior mathematics undergraduate).

Because of this, I have prepared a (perhaps excessively) extensive preliminary section and appendix. Furthermore, I may have overextended the thesis' scope, resulting in many incomplete sections in the rough draft. I was originally planning to cut out the preliminary section entirely, or perhaps place it some extended appendix. However, no prior honors thesis (to the best of my skimming) has ever used the appendix in such a manner, nor excluded so much introduction. Thus, since it is easier to cut than to add, I have left it all in.

Furthermore, the main results of the research connected to this thesis is still a work in progress — indeed, we have not resolved the main question! Because of this, I found it challenging to create some sort of cohesive presentation of the work besides an organized diary of failed attempts. Because of this, I recognize much work needs to be done to better “carve out” the original research from what is known to the field. At the same time, I feel there is merit in surveying all the developments pertaining this study, even if this dwarfs the original contributions.

DRAFT 2

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1 Introduction

We start with an informal introduction to quantum computing and lay some motivations.

Fault-tolerant quantum computing has the potential, through algorithms like Shor’s factoring algorithm [Sho97], to deliver possibly exponential speedups over classical computation. However, current Noisy Intermediate-Scale Quantum (NISQ) computers have too few qubits and too much noise to deliver practical benefit [Pre18].

Computation in the NISQ era is often modelled by a hybrid setup: on the quantum computer, a noisy initial state is prepared, some noisy gates are applied, and qubits are measured. A classical computer then processes these measurements, and can modify and repeat the quantum computation if necessary.

For the quantum computer, there exist a variety of computational models: The quantum Turing machine [Deu85], the quantum circuit model [Bar+95], adiabatic quantum computation [Far+01], and measurement-based models [RB01], among many others. Though these models are equivalent from a complexity viewpoint (informally, any problem that is efficiently solvable in one is efficiently solvable in the others) [RB01; Aha+08; Chi93], the underlying quantum properties used by each may differ significantly.

The measurement-based quantum computation (MBQC) model is of particular interest. This is because MBQC collapses any quantum circuit into measurements of a highly-entangled resource state that may be prepared in constant depth. This is quite shocking, as it means that any quantum computation may be done in constant depth. The cost of this model, however, is that measurements need to be adaptive. In other words, processing must be done on-the-fly to determine what qubits should be measured next and how.

Since measurement is irreversible, MBQC leads to a model of one-way quantum computation. In other words, as measurement cannot increase the entanglement of a system, all entanglement required must be initially present in the resource state. Informally then, entanglement is the resource consumed by this model of quantum computation. Thus, there is interest in determining what resource states are universal for quantum computation.

Graph states are a kind of resource state whose data corresponds to a simple, undirected graph. The comparability grid is a graph with vertices $V = \{(x, y) \mid x, y \in [n]\}$ and an edge between (x_1, y_1) and (x_2, y_2) if $x_1 \geq x_2$ and $y_1 \geq y_2$, excluding self-loops. It is an open problem whether the family of all comparability grids is universal for MBQC. There are implications for both possibilities. For example, if comparability grids are not universal, then unbounded rank width is not a sufficient condition for the universality of a family of graph states, since the family of all comparability grids have unbounded rank width.

In this thesis, we report on progress to prove and disprove the universality of comparability grids. Concretely, we give a set of conjectures for how one may be able to extract universal quantum computation using comparability grids as our resource state. In tandem to this, we outline various algorithms to verify small cases of these conjectures.

Readers with no prior exposure to quantum computation are encouraged to start at Appendix A. In Section 2, we introduce the necessary notation and background so that in Section 3 we may formally define the concepts in our problem statement. Then, Section 4 recalls some results regarding comparability grids. These theoretical results inspire the algorithmic approaches in Section 5, which in turn guide the results in Section 6.

2 Background

First some notation. Let $\mathbb{N} = \{1, 2, \dots\}$. For $k \in \mathbb{N}$, define $[k] := \{1, \dots, k\}$ and $[0] := \emptyset$. Let I be the identity matrix of any size and of any vector space, which will be clear from context. For a graph G , let $V(G)$ denote its vertex set and $E(G)$ its edge set. For any vector v with n entries, let v_i be its i th entry for all $i \in [n]$. For any matrix A with n rows and m columns, let $A_{i,j}$ be the entry at the i th row and j th column, for all $i \in [n], j \in [m]$.

We assume the reader is familiar with the mathematical foundations of quantum computing (e.g. at the level of an upper undergraduate quantum computing class). Please see Appendix A otherwise.

2.1 The ZX-Calculus

DEFINITION 2.1 The **ZX-Calculus** is a graphical language for representing quantum computation. Objects of the ZX-Calculus, called **ZX Diagrams**, are made up of two types of generators known as **spiders**: The **X-spider**, with n inputs and m outputs, is defined

$$n \text{ --- } \textcircled{\alpha} \text{ --- } m := |0\rangle^{\otimes m} \langle 0|^{\otimes n} + e^{i\alpha} |1\rangle^{\otimes m} \langle 1|^{\otimes n}. \quad (2.1)$$

We say that the above spider has a **phase** $\alpha \in [0, 2\pi)$. Similarly, the **Z-spider** is defined

$$n \text{ --- } \textcircled{\alpha} \text{ --- } m := |+\rangle^{\otimes m} \langle +|^{\otimes n} + e^{i\alpha} |-\rangle^{\otimes m} \langle -|^{\otimes n}. \quad (2.2)$$

In other words, the ZX-Calculus is a calculus (i.e. rules) for manipulating Z and X spiders.

REMARK 2.2 In the literature, the X-spider is often red and the Z-spider is often green, e.g. ● and ●. However, for accessibility reasons we choose to adopt the black-and-white ZX-Calculus color scheme instead.

DEFINITION 2.3 Write $a \propto b$ if $a = c \cdot b$, for some $c \in \mathbb{C}$. Furthermore, if D is a ZX diagram, write $\llbracket D \rrbracket$ to denote the matrix represented by D . To reduce notational clutter, we will use $\llbracket D \rrbracket$ only when it is necessary to distinguish a diagram from its matrix.

EXAMPLE 2.4 The standard quantum states can be written as spiders without inputs:

$$\textcircled{0} \text{ --- } = \sqrt{2} |+\rangle \propto |+\rangle, \quad \textcircled{\pi} \text{ --- } = \sqrt{2} |-\rangle \propto |-\rangle, \quad (2.3)$$

$$\textcircled{0} \text{ --- } = \sqrt{2} |0\rangle \propto |0\rangle, \quad \textcircled{\pi} \text{ --- } = \sqrt{2} |1\rangle \propto |1\rangle. \quad (2.4)$$

Single qubit gates can be written as spiders with 1 input and 1 output:

$$\text{---} \textcircled{\alpha} \text{ ---} = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{bmatrix} = e^{i\alpha/2} \begin{bmatrix} e^{-i\alpha/2} & 0 \\ 0 & e^{i\alpha/2} \end{bmatrix} = e^{i\alpha/2} R_z(\alpha) \quad (2.5)$$

$$\text{---} \textcircled{\alpha} \text{ ---} = \frac{1}{2} \begin{bmatrix} 1 + e^{i\alpha} & 1 - e^{i\alpha} \\ 1 - e^{i\alpha} & 1 + e^{i\alpha} \end{bmatrix} = e^{i\alpha/2} R_x(\alpha) \quad (2.6)$$

Thus $\text{---} \boxed{Z} \text{---} \propto \text{---} \textcircled{\pi} \text{---}$ and $\text{---} \boxed{X} \text{---} \propto \text{---} \textcircled{\pi} \text{---}$.

DEFINITION 2.5 Informally, the composition of spiders in ZX diagrams is similar to the composition of quantum gates in quantum circuits: Spiders connected via wires multiply, while spiders “stacked vertically” are tensored together.¹

DEFINITION 2.6 Here is an example of ZX composition: The Hadamard gate satisfies

$$\boxed{H} = e^{-i\pi/4} \text{---} \textcircled{\pi/2} \text{---} \textcircled{\pi/2} \text{---} \textcircled{\pi/2} \text{---}. \quad (2.7)$$

Due to this verbose representation, we shorthand $\text{---} \square \text{---} := \boxed{H}$ and call this the **H-box**.

DEFINITION 2.7 **Wires** or lines between spiders may “bend” in the following ways:

$$\text{X} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \text{C} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \quad \text{) = } [1 \quad 0 \quad 0 \quad 1]. \quad (2.8)$$

These are called “swap,” “cup,” and “cap,” respectively from left to right, and their composition is treated like the composition with a 2-qubit gate.

DEFINITION 2.8 We will be doing plenty of work with spiders moving forward, so we will set some conventions: The **arity** of a spider is the number of inputs and outputs it has. If a spider has phase 0, we say it is **phase-less**, and often do not bother writing the 0. Namely, $\bigcirc := \textcircled{0}$ and $\bullet := \textcircled{0}$.

DEFINITION 2.9 The ZX Calculus admits a finite set of **rewrite rules**, or graphical equivalences for rewriting ZX diagrams. These are summarized in Table 1. For further explanation of these rules and some history on their names, please see [Wet20].

So far, it is not clear why we would choose to use ZX-Calculus over either the algebraic or circuit representation of a quantum circuit. This becomes clearer when we look at Table 1. Namely, Table 1 tells us that there is a short, finite set of rules for rewriting any ZX diagram to another. Paired with the fact that there are only two types (three if you count the H-box) of objects being manipulated, this notation allows us to reason about quantum computation at a higher level compared to the matrix or circuit representations. The following lemma is an example of this.

LEMMA 2.10 (COLOR CHANGE LEMMA) Any rewrite identity in ZX holds if we replace Z-spiders by X-spiders and vice versa.

¹Formally, the category of ZX diagrams, $\mathbf{ZX}$, is a compact closed symmetric monoidal category. The objects are non-negative integers and the morphisms ZX diagrams. The composition of morphisms correspond to the composition of ZX diagrams. The tensor product on objects satisfies $n \otimes m := n + m$ and the tensor product of two ZX diagrams is given by “stacking” the diagrams on top of another. Due to the existence of the cup and cap, this is a compact closed category. Interested readers should see [AC07] for more details.

Name	Rewrite rule	Description
Spider fusion (f)	$ \begin{array}{c} n_1 \vdots \alpha \vdots m_1 \\ n_2 \vdots \beta \vdots m_2 \end{array} = + \vdots \begin{array}{c} \alpha + \beta \\ \vdots \\ \alpha + \beta \end{array} \vdots + \begin{array}{c} m_1 \\ \vdots \\ m_2 \end{array} $	Spiders of the same color connected by any number of wires fuse and their phases add.
Identity removal (id)	$ \text{---} \circ \text{---} = \text{---} $	Spiders with one input, one output, and phase 0 are the identity matrix and can be removed.
Hadamard cancellation (hh)	$ \text{---} \square \square \text{---} = \text{---} $	Two connected Hadamards cancel each other.
π -commutation (π)	$ \text{---} \pi \text{---} \alpha \vdots n = e^{i\alpha} \vdots n \text{---} -\alpha \text{---} \pi \text{---} $	A π phase copies through a spider of the opposite color and flips its phase.
State copy (cp)	$ \text{---} a\pi \text{---} \alpha \vdots n = e^{ia\alpha} \sqrt{2}^{-(n-1)} \vdots n \text{---} a\pi \text{---} $	A π phase copies through a spider of the opposite color and flips its phase.
Color change (cc)	$ n \vdots \alpha \vdots m = n \vdots \alpha \vdots m $	Commuting Hadamards through a spider changes the color.
Bi-algebra (b)	$ n \vdots \alpha \vdots m = \sqrt{2}^{(n-1)(m-1)} n \vdots \alpha \vdots m $	An adjacent phase 0 X and Z spider can commute past one another to create a complete bipartite graph with sides of the opposite color.
Hopf rule (hf)	$ \text{---} \circ \circ \text{---} = \frac{1}{2} \text{---} \circ \text{---} $	If phase 0 spiders of opposite color are connected by multiple wires, we may remove the wires pairwise.

Table 1: ZX-Calculus Rewrite Rules

Proof. Consider a ZX diagram identity $D = D'$. For both: Apply (cc) on all spiders to swap all colors. All inputs and outputs connected to a spider gain a H-box, which are the same wires in D and D' . All wires between two spiders gain two H-boxes, which by (hh) cancel. Thus the color-swapped D is equivalent to the color-swapped D' . $\square$

THEOREM 2.11 Any ZX diagram is topologically invariant; only connectivity matters.

Proof. The proof surmounts to a tedious enumeration of all possible rearrangements of the same quantum circuit [CK17]. Like in quantum circuits, we often write ZX diagrams where inputs come from the left and outputs go to the right. This is simply convention. More formally, it suffices to maintain an ordered list of wires considered the input(s) and output(s). $\square$

EXAMPLE 2.12 The CNOT gate has the following ZX representation:

$$\begin{array}{c} \bullet \\ | \\ \oplus \end{array} = \begin{array}{c} \circ \\ | \\ \bullet \end{array}. \quad (2.9)$$

What Theorem 2.11 says is that

$$\left[\begin{array}{c} \circ \\ | \\ \bullet \end{array} \right] = \left[\begin{array}{c} \bullet \\ | \\ \oplus \end{array} \right] = \left[\begin{array}{c} \bullet \\ | \\ \oplus \end{array} \right]. \quad (2.10)$$

LEMMA 2.13 The ZX-Calculus is exactly universal for quantum computation², up to some complex scalar factor.

Proof. By Example 2.4 and Euler decomposition (Theorem A.12), we have that every single-qubit gate can be represented in the ZX-Calculus, up to some complex scalar factor. Example 2.12 gives us a representation of CNOT, a two-qubit entangling gate in the ZX-Calculus. By Theorem 1.3 of [BB01], arbitrary single-qubit gates with any entangling 2-qubit gate are exactly universal for quantum computation. $\square$

THEOREM 2.14 Any $c \in \mathbb{C}$ can be written as a ZX diagram.

Proof. Notice³

$$\left[\begin{array}{c} \pi \\ \theta \end{array} \right] = e^{i\theta}, \quad \left[\begin{array}{c} \alpha \end{array} \right] = 1 + e^{i\alpha}, \quad \left[\begin{array}{c} \bullet \bullet \\ \bullet \bullet \end{array} \right] = \frac{1}{2}, \quad \left[\begin{array}{c} \circ \end{array} \right] = 2. \quad (2.11)$$

Then

$$\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2} = \frac{1}{2} (e^{-i\theta} (1 + e^{2i\theta})). \quad (2.12)$$

²See Definition 3.2 for a clarification of what exactly universal means.

³Here we use the fact that composition between ZX diagrams that are scalars is regular multiplication. This is a subtle implication of 2.5.

Thus we can write $\cos(\theta)$ as a ZX diagram. By the Archimedean property, for all $r \in \mathbb{R}$, there exists $k \in \mathbb{N}$ such that $2^k > |r|$. So we can write

$$r = 2^k \cos(\theta), \quad \theta = \arccos\left(\frac{r}{2^k}\right). \quad (2.13)$$

Then any real number can be written as a ZX diagram. Now, for any $c \in \mathbb{C}$, use the polar coordinates $c = \rho \cdot e^{i\theta}$ to obtain a ZX representation of c . Note that for $c = 0$, $\llbracket \pi \rrbracket = 1 - 1 = 0$. $\square$

THEOREM 2.15 The ZX-Calculus is exactly universal for quantum computation.

Proof. This follows by Theorem 2.14 and Lemma 2.13. $\square$

Furthermore, it is a well-known result [JPV20] that the rules in Table 1 are complete for quantum computation. That is, any identity that can be shown through the matrix representation can also be shown through ZX-Calculus. Equivalently, if two diagrams D, D' represent the same quantum computation, the diagrams can be transformed to one another using only the rules in Table 1. In fact, we may largely ignore scalar factors during this rewrite process. This is formalized below:

LEMMA 2.16 We may ignore scalar factors during ZX rewrites, provided that the desired scalar is restored at the end. This gives us the computation up to a global phase.

Proof. Given a quantum computation as a unitary U , we can obtain a diagram D where $\llbracket D \rrbracket = cU$, where $c \in \mathbb{C}$ is some non-zero scalar factor. We then have $\llbracket D \rrbracket \llbracket D \rrbracket^\dagger = c\bar{c}I$. Then

$$\frac{1}{|c|} \llbracket D \rrbracket = \frac{c}{|c|} U, \quad (2.14)$$

so we can recover U up to a global phase. $\square$

Computationally, it is often convenient to keep track of the product of all scalar factors in a diagram as a *number* rather than as a ZX diagram. We can use this extraneous data structure precisely because of Theorem 2.14, and as such we will recover the computation's global phase as well (unlike Lemma 2.16).

2.2 Clifford Circuits

DEFINITION 2.17 A **Clifford circuit** is a quantum circuit made of **Clifford gates**:⁴

$$\text{CNOT} := \begin{array}{c} \text{---} \circ \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array}, \quad H := \text{---} \square \text{---} = \begin{array}{c} \text{---} \circ \text{---} \\ | \\ \text{---} \bullet \text{---} \end{array} \begin{array}{c} \text{---} \pi/2 \text{---} \\ | \\ \text{---} \pi/2 \text{---} \end{array}, \quad S := \text{---} \pi/2 \text{---}. \quad (2.15)$$

Furthermore, let H and S be known as the **local Clifford gates**.

⁴This is not the typical formal definition of Clifford circuits. However, we will soon show that this definition is equivalent to the usual one, i.e. as the circuits that normalize the Pauli group.

DEFINITION 2.18 A **Clifford ZX diagram** is a ZX diagram made up only of spiders whose phases are multiples of $\frac{\pi}{2}$. Furthermore, say a ZX diagram D is **unitary** if $\llbracket D \rrbracket = cU$ for some $c \in \mathbb{C}$ and unitary matrix U .

THEOREM 2.19 Clifford circuits are precisely the unitary Clifford ZX diagrams. That is, every Clifford circuit can be written as a unitary Clifford ZX diagram and vice versa.

Proof, part 1. It is clear that any Clifford circuit is a Clifford ZX diagram, since the diagrams of the Clifford gates only involve spiders with phases multiples of $\frac{\pi}{2}$. The reverse direction, however, will take more work. $\square$

To show the reverse direction, we construct an algorithm from Chapter 5 of [KW24] that allows us to convert any unitary Clifford ZX diagram into a Clifford circuit.

DEFINITION 2.20 A **Hadamard wire** is a wire with a single Hadamard, i.e. $\text{---}\square\text{---}$. To further disambiguate this from $\text{---}\bigcirc\text{---}$, we may write $\text{---}\square\text{---}$ in place of $\text{---}\square\text{---}$.

DEFINITION 2.21 A ZX diagram is **graph-like** when:

- (a) Every spider is a Z-spider.
- (b) Spiders are only connected via Hadamard edges.
- (c) There are no self loops or parallel / duplicate edges.
- (d) Every Z-spider is connected to at most one input and at most one output via a regular edge.
- (e) Every input and output is connected to a Z-spider.

THEOREM 2.22 Every ZX diagram can be efficiently rewritten as a graph-like diagram.

Proof. Let D be an arbitrary ZX diagram. Run the following algorithm:

1. Convert all X-spiders to Z-spiders via (cc) .
2. Cancel adjacent H with (hh) .
3. Fuse all connected spiders via (cc) until this cannot be done.
4. Remove parallel Hadamard edges via:

$$\text{---}\bigcirc\text{---}\square\text{---}\bigcirc\text{---}\beta \stackrel{(cc)}{=} \text{---}\square\text{---}\alpha\text{---}\square\text{---}\beta \stackrel{(hf)}{=} \text{---}\square\text{---}\alpha\text{---}\bigcirc\text{---}\beta \stackrel{(cc)}{=} \text{---}\alpha\text{---}\beta. \quad (2.16)$$

5. Remove self-loops via:

$$\text{---}\bigcirc\text{---}\alpha \stackrel{(id)}{=} \text{---}\bigcirc\text{---}\alpha \stackrel{(f)}{=} \text{---}\alpha\text{---}, \quad (2.17)$$

$$\text{---}\bigcirc\text{---}\alpha \stackrel{(f)}{\propto} \text{---}\bigcirc\text{---}\alpha + \pi \stackrel{(hf)}{\propto} \text{---}\alpha + \pi\text{---}. \quad (2.18)$$

6. Introduce Z-spiders and Hadamards using (id) and (hh) to ensure every input and output is connected to a Z-spider and no Z-spiders are connected to multiple inputs or multiple outputs.

After this, all spiders are Z-spiders. All edges between spiders are Hadamard edges, else the spiders would have been fused together. There is at most one edge between any two spiders, and there are no self-loops. Step 6 ensures (e) is satisfied. $\square$

Note that the above procedure applies to any ZX diagram, not just Clifford ones. As it turns out, when our starting ZX diagram is Clifford, we can simplify the diagram even further. To see how, we introduce the notion of a graph state and some operations on them.

DEFINITION 2.23 Say a spider in a ZX diagram is **internal** if it is not directly connected to an input or output wire, but it has a connection via other spiders to an input or output wire. Conversely, a spider is a **boundary** spider if it is directly connected to an input or output wire.

DEFINITION 2.24 For a simple, undirected graph $G = (V, E)$, the **graph state** $|G\rangle$ of G is defined

$$|G\rangle := \prod_{(v,w) \in E} CZ_{v,w} |+\rangle^{\otimes |V|}. \quad (2.19)$$

That is, prepare the $|+\rangle^{\otimes |V|}$ state and apply a CZ gate between every pair of adjacent vertices in G . If D is a ZX diagram and graph state, we write $G(D) = (V, E)$ to refer to the graph. Note that in the ZX Calculus,

$$|+\rangle \propto \text{---} \bigcirc \text{---}, \quad CZ = \begin{array}{c} \text{---} \bigcirc \text{---} \\ | \\ \text{---} \bigcirc \text{---} \end{array}. \quad (2.20)$$

EXAMPLE 2.25 The triangle graph has the following circuit and ZX graph state:

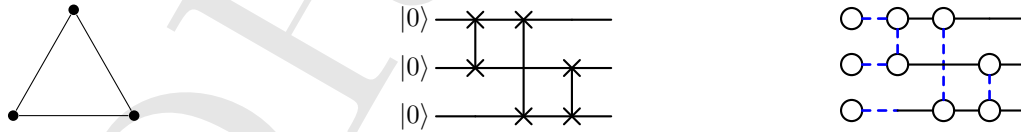


Figure 1: The triangle graph and its circuit and ZX graph state.

DEFINITION 2.26 A **graph state with local Cliffords (GSLC)** is a graph state to which some single-qubit Clifford unitaries have been applied on its outputs.

THEOREM 2.27 Every Clifford ZX diagram can be written in GSLC form.

To prove Theorem 2.27, we need to show that all internal spiders in a Clifford ZX diagram in graph-like form can be removed, perhaps by changing the phases of some of the boundary spiders and adding local Cliffords. To do this we need the following graph operations.

DEFINITION 2.28 Let G be a graph and $u \in V(G)$. The **local complementation** of G around u , written $G \star u$, is the graph obtained by complementing the subgraph induced by the neighbors of u . For distinct $v, w \in V(G)$, let $e_{v,w}$ be the edge between them. Then

$$e_{v,w} \in E(G \star u) \iff \begin{cases} e_{v,w} \notin E(G) & \text{if } v, w \in N_G(u), \\ e_{v,w} \in E(G) & \text{otherwise.} \end{cases} \quad (2.21)$$

LEMMA 2.29 Let D be a graph-state ZX diagram, and u some vertex in $G(D)$. Then we can obtain a diagram D' with $G(D') = G(D) \star u$ by applying a $R_X(-\pi/2)$ gate on the spider corresponding to u and $R_Z(\pi/2)$ on all spiders in $N_{G(D)}(u)$. Call this rule (lc) :

$$\quad (2.22)$$

Thus by applying local Cliffords we can local complement a graph state.

Proof. This was originally proved in Theorem 2 of [09]. The proof surmounts to a clever decomposition of the H-box as Z and X-spiders, which after some application of the rules in Table 1 gives the desired result. $\square$

LEMMA 2.30 The following holds:

$$\quad (2.23)$$

Here, the $\star$ above $(\pm \frac{\pi}{2})$ indicates a local complementation at this vertex.

Proof. We provide a proof sketch: Let u denote the vertex we local complement around. First, push out all the phases via (f) to obtain a graph state where all the vertices are phase-free Z-spiders. Apply local complementation in reverse around u . Then u will have $(\pm \frac{\pi}{2}) - (\pm \frac{\pi}{2}) = \bigcirc$ —dangling from it. Apply (cp) to push the $\bigcirc$ — through and disconnect u . This gives the RHS. For a step-by-step proof, see Appendix B.3 of [Dun+20]. $\square$

DEFINITION 2.31 Let G be a graph, $u, v \in V(G)$, and $e_{uv} \in E(G)$. The pivot of G along uv , denoted $G \wedge uv$, is defined as

$$G \wedge uv := G \star u \star v \star u \quad (2.24)$$

or equivalently

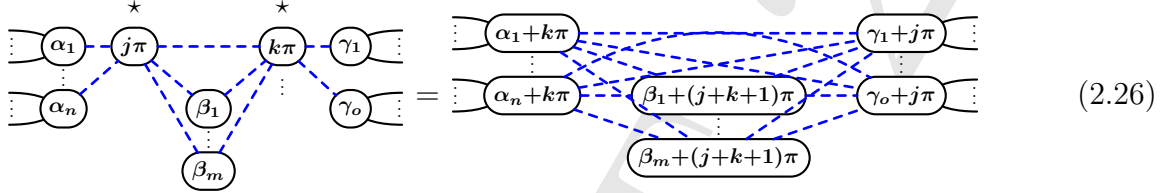
$$G \wedge uv = G \star v \star u \star v. \quad (2.25)$$

Proof. This is a standard proof by cases, seen for example in Definition 4.3 of [Dun+20]. $\square$

LEMMA 2.32 Given a graph G , we can pivot around uv for $u, v \in V(G)$ by applying Hadamard on u and v and Z on all vertices in $N_G(u) \cap N_G(v)$, inducing a swap gate between u and v .

Proof. The proof is surprisingly involved and can be found in Proposition 3.1 of [DP14]. $\square$

LEMMA 2.33 The following holds:

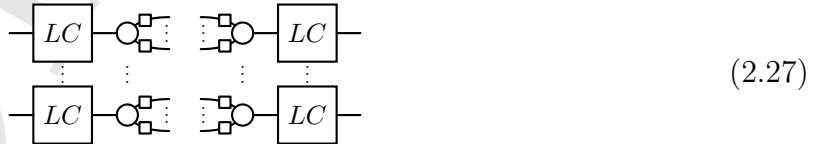


Note that the vertices $\beta_1, \dots, \beta_m$ may have other wires, but we suppress them here to reduce clutter.

Proof. We provide a proof sketch: Let u denote the vertex with phase $j\pi$ and v the vertex with phase $k\pi$. First, push out all the phases via (f) to obtain a graph state where all the vertices are phase-free Z-spiders. Apply pivoting on uv . Then u will have $(k\pi)$ — $\square$ dangling from it (note the change from j to k due to the pivoting's swap gate). Apply (cc) and (cp) to obtain the RHS after fusing with (f) . For a step-by-step proof, see Appendix B.3 of [Dun+20]. $\square$

Proof of Theorem 2.27. First note that by Theorem 2.11, we can topologically morph the inputs of the Clifford circuit to face rightwards, like outputs, and that this has no effect on the operation represented by the diagram. Now, start by converting a Clifford ZX diagram to graph-like form. Apply Lemma 2.30 and 2.33 to remove all internal spiders from the diagram, perhaps appending local Clifford gates to the output wires. This gives a circuit in GSLC form. $\square$

LEMMA 2.34 The GSLC form of a CLifford ZX diagram may be written as follows:



Here $\boxed{LC}$ refers to circuits of the form $H^i S^j$ for some $i \in [0, 1]$ and $j \in [0, 1, 2, 3]$.

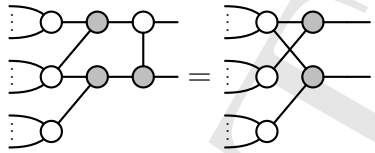
Proof. First note that we may topologically morph the GSLC state such that input wires face left and output wires face right. Now, note that the local Clifford generated must be of the form $H^i S^j$. Namely, when applying Lemma 2.30, we only modify the phase of boundary spiders. When we apply Lemma 2.33, we may add a Hadamard gate to an input or output wire. Since two Hadamards cancel, each input or output wire has at most one Hadamard. Since the phase of each boundary spider is a multiple of $\frac{\pi}{2}$ modulo 2π , unfusing the phase requires at most three S gates. $\square$

LEMMA 2.35 Let U be a Clifford unitary. Then U can be written as a circuit consisting of 8 layers made out of only the following gates in order:

$$H \quad S \quad CZ \quad CNOT \quad H \quad CZ \quad S \quad H \quad (2.28)$$

Proof. This proof is taken from Theorem 5.3.11 of [KW24]. We provide a proof sketch: Start from Lemma 2.34. This explains the outer H and S layers. Now, apply (f) in reverse to obtain a CZ , representing an edge between two vertices both on the left or both on the right. This explains the two CZ layers. Then, apply (cc) to obtain the H layer in between $CNOT$ and CZ .

What remains is a bipartite graph whose left vertices are phase-less Z -spiders and right vertices phase-less X -spiders. As it turns out, because our diagram is unitary, this bipartite graph can be decomposed into a circuit made only of $CNOT$ gates via the following identity:

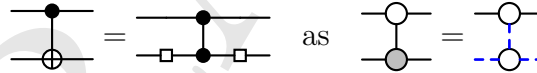


$$(2.29)$$

The proof of this surmounts to book-keeping and a clever use of (b) . Please see Lemma 4.2.11 of [KW24] for a step-by-step proof. $\square$

Proof of Theorem 2.27 and Proof of Theorem 2.19, part 2. Given a unitary Clifford ZX diagram, do the following:

1. Convert it to graph-like form, using Theorem 2.22.
2. Use Lemma 2.30 and Lemma 2.33 to remove internal spiders and bring the diagram to GSLC form.
3. Use Lemma 2.35 to obtain a Clifford circuit for the diagram, by noting that



$$(2.30)$$

Since we have provided an injective map in both directions, the proof of Theorem 2.19 is complete. $\square$

COROLLARY 2.36 Any n -qubit Clifford unitary can be represented by an equivalent circuit in at most $4n(n + 3/2) - 1$ Hadamard, S , and $CNOT$ gates.

Proof. Look at Lemma 2.35. Each Hadamard layer has at most n gates, and there are three of those. Each S gate layer has at most $3n$ gates, and there are two of them. A circuit of CZ gates can have at most one CZ gate for every pair of qubits, as CZ gates commute past each other. Thus the CZ layer has at most $n(n - 1)/2$ gates. CZ s are made up of one $CNOT$ and two Hadamards, so this contributes $3 \cdot n(n - 1)/2$ gates. Finally, the $CNOT$ circuit is essentially the number of distinct invertible maps from $\mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$. It is left as an exercise to the reader to see that there are at most $n^2 - 1$ such maps. Thus, in total at most

$$9n + 2 \cdot 3 \cdot \frac{n(n - 1)}{2} + n^2 - 1 = 4n(n + 3/2) - 1 \quad (2.31)$$

gates are used. $\square$

THEOREM 2.37 Let U be a q -qubit Clifford unitary with N gates. We can rewrite this as a ZX diagram in GSLC form in $O(Nq^2)$ time, if $N \geq q$. Further, we can calculate any amplitude

$$\langle x_1, \dots, x_n | U | 0, \dots, 0 \rangle \quad (2.32)$$

in $O(Nq^2)$ time. This is practically equivalent to the **Gottesman-Knill theorem**. If U is already available in GSLC form, this can be done in $O(q^3)$ time.

Proof. The first comes from analyzing the runtime of the algorithm in Theorem 2.19's part 2 proof. The second comes from the fact that, a post-selection of outcomes $|x_1, \dots, x_n\rangle$ is equivalent to “plugging” the output wires of a GSLC diagram with the corresponding spiders. The third claim follows from a similar observation. Please see Proposition 5.4.4 of [KW24] for a more detailed proof. $\square$

2.3 The Stabilizer Formalism

When we view the Clifford circuits of the previous section more algebraically, we find interesting connections to the following objects:

DEFINITION 2.38 The **Pauli matrices** or **Pauli gates** refer to the following gates:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \quad (2.33)$$

DEFINITION 2.39 Say a matrix A is a **stabilizer** of the state $|\psi\rangle$ when $|\psi\rangle$ is a $+1$ eigenvector of A , i.e. $A|\psi\rangle = |\psi\rangle$.

DEFINITION 2.40 The n -qubit Pauli group is defined as follows:

$$\mathcal{P}_n := \{i^k P_1 \otimes \dots \otimes P_n \mid k \in \{0, 1, 2, 3\}, P_j \in \{I, X, Y, Z\}\}.$$

Call the elements of $\mathcal{P}_n$ as **Pauli strings**, written as vectors of length n :

$$\vec{P} := i^k P_1 \otimes \dots \otimes P_n.$$

Proof. We need to show that this is indeed a group. Associativity follows from matrix multiplication. I is the identity. Each Pauli matrix is its own inverse, i.e. $X^2 = Y^2 = Z^2 = 1$, and it is easy to see that the inverses of $i^k P$ exist for all k and P . Finally, since two Pauli matrices $P_1 \neq P_2$ anti-commute, i.e. $P_1 P_2 = -P_2 P_1$, an exhaustive multiplication table shows the group is closed and finite. $\square$

THEOREM 2.41 For any Clifford unitary U and Pauli string $\vec{P}$ of compatible dimensions, there exists a pauli string $\vec{Q}$ such that $U\vec{P}U^\dagger = \vec{Q}$.

Proof. It suffices to check that, when we commute a basis of the Pauli strings through the Hadamard, S, and CNOT gate, we will end up with another Pauli string. Notice that $\{X, Z\}$ is a basis for $\mathcal{P}_1$, up to scalar factors. Now note that

$$\text{---} \circ \pi \text{---} \square \text{---} \stackrel{(cc)}{=} \text{---} \square \text{---} \circ \pi \text{---}, \quad \text{---} \circ \pi \text{---} \square \text{---} \stackrel{(cc)}{=} \text{---} \square \text{---} \circ \pi \text{---}. \quad (2.34)$$

For the S gate,

$$\text{---} \circ \pi \text{---} \circ \frac{\pi}{2} \text{---} \stackrel{(f)}{=} \text{---} \circ \frac{\pi}{2} \text{---} \circ \pi \text{---}, \quad \text{---} \circ \pi \text{---} \circ \frac{\pi}{2} \text{---} \stackrel{\begin{smallmatrix} (\pi) \\ (f) \end{smallmatrix}}{=} \text{---} \circ \frac{\pi}{2} \text{---} \circ \pi \text{---} \circ \pi \text{---}. \quad (2.35)$$

For the CNOT gate,

$$\begin{array}{ccc} \text{---} \circ \pi \text{---} \circ \text{---} & \stackrel{(f)}{=} & \text{---} \circ \text{---} \circ \pi \text{---} \\ \text{---} \circ \text{---} & & \text{---} \circ \text{---} \end{array}, \quad \begin{array}{ccc} \text{---} \circ \pi \text{---} \circ \text{---} & \stackrel{\begin{smallmatrix} (\pi) \\ (f) \end{smallmatrix}}{=} & \text{---} \circ \text{---} \circ \pi \text{---} \\ \text{---} \circ \text{---} & & \text{---} \circ \pi \text{---} \end{array}, \quad (2.36)$$

$$\begin{array}{ccc} \text{---} \circ \text{---} \circ \text{---} & \stackrel{\begin{smallmatrix} (\pi) \\ (f) \end{smallmatrix}}{=} & \text{---} \circ \text{---} \circ \pi \text{---} \\ \text{---} \circ \pi \text{---} \circ \text{---} & & \text{---} \circ \pi \text{---} \end{array}, \quad \begin{array}{ccc} \text{---} \circ \pi \text{---} \circ \text{---} & \stackrel{(f)}{=} & \text{---} \circ \text{---} \circ \pi \text{---} \\ \text{---} \circ \pi \text{---} & & \text{---} \circ \pi \text{---} \end{array}. \quad \square$$

DEFINITION 2.42 For a group $\mathcal{S} \subseteq \mathcal{P}_n$, define the **stabilizer subspace**⁵ of $\mathcal{S}$ as

$$\text{Stab}(\mathcal{S}) := \{|\psi\rangle \mid \vec{P}|\psi\rangle = |\psi\rangle, \forall \vec{P} \in \mathcal{S}\}. \quad (2.37)$$

If $\text{Stab}(\mathcal{S}) \neq \{0\}$, call $\mathcal{S}$ a **stabilizer group**.

Proof. If A, B are stabilizers of $|\psi\rangle$, then $AB|\psi\rangle = A|\psi\rangle = |\psi\rangle$ so AB is also a stabilizer. Furthermore,

$$A|\psi\rangle = |\psi\rangle \implies |\psi\rangle = A^{-1}|\psi\rangle$$

so A^{-1} is also a stabilizer of $|\psi\rangle$. Thus $\mathcal{S}$ is a subgroup. On the other hand, if two states $|\psi\rangle$ and $|\phi\rangle$ are stabilized by A , then

$$A(\lambda|\psi\rangle + \mu|\phi\rangle) = \lambda A|\psi\rangle + \mu A|\phi\rangle = \lambda|\psi\rangle + \mu|\phi\rangle, \quad (2.38)$$

for any $\lambda, \mu \in \mathbb{C}$. Since 0 is trivially in every $\text{Stab}(\mathcal{S})$, this is indeed a subspace. $\square$

LEMMA 2.43 If a subgroup $\mathcal{S} \subseteq \mathbb{P}_n$ is a stabilizer subgroup, then:

1. $-I \notin \mathcal{S}$,
2. $\vec{P}^2 = I$ for all $\vec{P} \in \mathcal{S}$, and
3. $\mathcal{S}$ is commutative.

⁵This is a strictly more restrictive definition of stabilizer subspaces and stabilizer groups than the definitions found in algebra, for instance. However, since we are only dealing with pure states, and are only concerned with the stabilizer subspaces of $\mathcal{P}_n$, this definition suffices.

Proof. If $-I \in \mathcal{S}$, then all $|\psi\rangle \in \text{Stab}(\mathcal{S})$ has $|\psi\rangle = -I|\psi\rangle = -|\psi\rangle$ such that $\text{Stab}(\mathcal{S}) = \{0\}$, a contradiction. An exhaustive multiplication table for $\mathcal{P}_1$ reveals that if $\vec{P}^2 \neq I$, then $\vec{P}^2 = -I$ so again $\text{Stab}(\mathcal{S}) = \{0\}$. Thus, each $P \in \mathcal{S}$ must have $P^2 = Q^2 = I$. Then $P^\dagger = P$, since $P^{-1} = P$ and its phase cannot be $\pm i$. Then for any $P, Q \in \mathcal{S}$, $PQ = (PQ)^\dagger = Q^\dagger P^\dagger = QP$. $\square$

THEOREM 2.44 This is Theorem 1 of [AG04]. Given an n -qubit state $|\psi\rangle$, the following are equivalent:

1. $|\psi\rangle$ can be obtained from $|0\rangle^{\otimes n}$ by CNOT, H, and S gates only.
2. $|\psi\rangle$ can be obtained from $|0\rangle^{\otimes n}$ by CNOT, H, S, and measurement gates only.
3. $|\psi\rangle$ is stabilized by exactly 2^n Pauli operators.
4. $|\psi\rangle$ is uniquely determined by $\text{Stab}(|\psi\rangle)$.

Thus, we can call any circuit consisting entirely of Clifford gates and measurements a **stabilizer circuit**, and any state obtainable by applying a stabilizer circuit to $|0\rangle^{\otimes n}$ a **stabilizer state**. Equivalently, a stabilizer state is a state $|\psi\rangle \in \mathbb{C}^{2^n}$ such that $\vec{P}|\psi\rangle = |\psi\rangle$ for all $\vec{P} \in \mathcal{S} \subseteq \mathcal{P}_n$.

Proof. A concise proof is available in [AG04], which itself relies on the Gottesman-Knill theorem (Theorem 2.37), whose ZX-proof can be found in Chapter 5 of [KW24] and non-ZX-proof in the original paper [Got98]. $\square$

LEMMA 2.45 Note that $|\mathcal{P}_n| = 4^{n+1}$. Furthermore, the number of stabilizer states on n qubits is:

$$2^n \prod_{i=0}^{n-1} (2^{n-i} + 1). \quad (2.39)$$

Proof. The first claim follows almost immediately from Definition 2.40. By Theorem 2.44, to obtain the second claim we can equivalently count the number of ordered generating sets for stabilizer groups, divided by the number of generating sets for a given stabilizer. This is the proof used in Proposition 2 of [AG04]. $\square$

Informally, what this stabilizer formalism tells us is that states obtainable by applying Clifford circuits to $|0\rangle^{\otimes n}$ are precisely the pure stabilizer states. This is significant for a whole host of reasons, not the least of which is the tableau formalism below. This tableau formalism essentially tells us that we can keep track of a Clifford computation by keeping track of changes to the stabilizer of the state.

DEFINITION 2.46 The tableau of a state $|\psi\rangle$ is a matrix consisting of binary variables x_{ij} and z_{ij} for $i \in [2n]$ and $j \in [n]$, and r_i for all $i \in [2n]$:

$$\left[\begin{array}{ccc|ccc|c} x_{11} & \cdots & x_{1n} & z_{11} & \cdots & z_{1n} & r_1 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{n1} & \cdots & x_{nn} & z_{n1} & \cdots & z_{nn} & r_n \\ \hline x_{(n+1)1} & \cdots & x_{(n+1)n} & z_{(n+1)1} & \cdots & z_{(n+1)n} & r_{n+1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots & \vdots \\ x_{(2n)1} & \cdots & x_{(2n)n} & z_{(2n)1} & \cdots & z_{(2n)n} & r_{2n} \end{array} \right] \quad (2.40)$$

where

$$x_{ij} = \begin{cases} 1 & \text{if } P_{ij} = X \text{ or } P_{ij} = Y, \\ 0 & \text{otherwise,} \end{cases} \quad (2.41)$$

$$z_{ij} = \begin{cases} 1 & \text{if } P_{ij} = Y \text{ or } P_{ij} = Z, \\ 0 & \text{otherwise,} \end{cases} \quad (2.42)$$

$$r_i = \begin{cases} 1 & \text{if } R_i \text{ has negative phase,} \\ 0 & \text{otherwise.} \end{cases} \quad (2.43)$$

THEOREM 2.47 Using the tableau representation, we can efficiently simulate CNOT, H, and S gates. This is an alternative method of simulation compared to Theorem 2.37.⁶

Proof. The proof surmounts to a clever bookkeeping of the stabilizers of a state as Clifford gates are applied to it. This proof can be found in [AG04]. Here we provide some intuition for why this is the case. The key observation is that each Pauli gate and I can be represented by two classical bits (x, z) :

$$\begin{array}{c|cc} P & x & z \\ \hline I & 0 & 0 \\ X & 1 & 0 \\ Z & 0 & 1 \\ Y & 1 & 1 \end{array} \quad (2.44)$$

Note that we represent Y as $(1, 1)$ even though $Y = iXZ$. This is due to Lemma 2.43; each Pauli stabilizer cannot have a phase of $\pm i$, but may have a sign ± 1 . Thus, for a Pauli string $\vec{P}$ of length n , we can capture its data using two bits for each of the n qubits, plus an extra bit r for the overall sign. The tableau has $2n$ rows for bookkeeping and algorithmic reasons.⁷ $\square$

This algorithm may be improved further. See [SSY25] for such an example.

⁶As it turns out, both methods of simulation, i.e. the ZX-Calculus GSLC algorithm and the tableau representation, will be used in this thesis.

⁷The first n rows keep track of the *de-stabilizers* of the state, while the next n rows keep track of the stabilizers. See [AG04] for more details.

3 Measurement-Based Quantum Computation

3.1 Universality

In this thesis, we are primarily concerned with the universality of the comparability grid for measurement-based quantum computation. It follows that we need a precise definition of universality to formally state our problem.

DEFINITION 3.1 For every $n \in \mathbb{N}$, Let $SU(n)$ denote the special unitary group under matrix multiplication consisting of $n \times n$ unitary matrices with determinant 1. It is a normal subgroup of the unitary group $U(n)$, consisting of $n \times n$ unitary matrices.

DEFINITION 3.2 A set of gates $\mathcal{S}$ is (efficiently) **universal** if any $U \in SU(2^n)$, for any $n \in \mathbb{N}$, that can be generated (in $\text{poly}(n)$ gates) by an arbitrary gate set $\mathcal{S}'$, can also be generated by the gate set $\mathcal{S}$ (in $\text{poly}(n)$ gates) [Nes+07]. We say a set of gates $\mathcal{S}$ is d -qubit universal if the above holds for $U \in SU(2^d)$.

In this thesis we will only focus on universality, not efficient universality. Thus, it may take exponential or more gates to simulate certain unitaries U . Also, in practice, the application of U up to some measure and error ϵ suffices. This is called **approximate universality**. This thesis will focus solely on **exact universality**, where we desire no error, rather than approximate universality. Henceforth, all references to universality will refer to exact, and not necessarily efficient, universality.

Furthermore, note that definition 3.2 is only concerned with implementing arbitrary unitaries, not necessarily arbitrary computation. That is, it does not specify whether the inputs and outputs are classical or quantum in nature. This matters because $|0\rangle$ and $-|0\rangle$ have the same measurement distribution and classical output, but $|0\rangle \neq -|0\rangle$. This difference is called **global phase**, i.e. a scalar $e^{i\theta}$ multiplied to an entire state.

DEFINITION 3.3 We use the following types of **universal computation** from [Nes+07]:

- **CC-universal**: For every $U \in SU(n)$, the model can reproduce the distribution of local measurements performed on $U|0^n\rangle$.
- **QC-universal**: Given an arbitrary n -qubit state $|\phi\rangle$, for every $U \in SU(2^n)$, the model can reproduce the distribution of local measurements performed on $U|\phi\rangle$.
- **CQ-universal**: For every $U \in SU(2^n)$, the model can produce the quantum state $|\phi\rangle = U|0^n\rangle$.
- **QQ-universal**: Given an arbitrary n -qubit state $|\phi\rangle$, for every $U \in SU(2^n)$, the model can produce the quantum state $U|\phi\rangle$.

These distinctions allow us to more finely categorize universality. For example, we may restate Lemma 2.16 as follows: Ignoring scalar factors during ZX rewrites, provided that the desired scalar is restored at the end, makes ZX-Calculus at most QC-universal, while keeping track of scalars throughout gives QQ-universality.

The first three CNOTs are equivalent to a swap gate. Thus

$$\begin{array}{c} H|\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \bullet \\ \oplus \\ \bullet \\ \oplus \end{array} \begin{array}{c} \oplus \\ \bullet \\ \oplus \\ \oplus \end{array} \begin{array}{c} \bullet \\ \bullet \\ \bullet \\ \bullet \end{array} \begin{array}{c} \boxed{\text{meter}} \\ \\ \\ \end{array} = \begin{array}{c} |+\rangle \\ H|\phi\rangle \end{array} \begin{array}{c} \bullet \\ \oplus \end{array} \begin{array}{c} \boxed{\text{meter}} \\ \\ \end{array} \quad (3.4)$$

The final state before measurement is

$$\text{CNOT}(|+\rangle \otimes H|\phi\rangle) = \frac{1}{\sqrt{2}} (|0\rangle \otimes H|\phi\rangle + |1\rangle \otimes XH|\phi\rangle). \quad (3.5)$$

In other words, a measurement of the first qubit in the **computational basis** ($\{|0\rangle, |1\rangle\}$) will tell us whether we have a $H|\phi\rangle$ or $XH|\phi\rangle$ on the output wire.

Now, suppose our intention in the above circuit was to apply a Hadamard gate to $|\phi\rangle$. Then there is a 0.5 probability that we apply XH instead of H , which is indicated when we measure a 1. How can we get rid of this additional X ?

EXAMPLE 3.6 Define the **teleportation gadget** as the following circuit.

$$\begin{array}{c} |\phi\rangle \\ |+\rangle \end{array} \begin{array}{c} \bullet \\ \bullet \end{array} \begin{array}{c} \boxed{R_Z(\theta)} \\ \boxed{H} \end{array} \begin{array}{c} \boxed{\text{meter}} \\ \\ \end{array} = \begin{array}{c} |+\rangle \\ HR_Z(\theta)|\phi\rangle \end{array} \begin{array}{c} \bullet \\ \oplus \end{array} \begin{array}{c} \boxed{\text{meter}} \\ \\ \end{array}. \quad (3.6)$$

Let $m \in \{0, 1\}$ be the measurement outcome of measuring in the computational basis. Then we apply $X^m HR_Z(\theta)$ to $|\phi\rangle$.

Proof. We show that the RHS circuit is indeed the LHS circuit. First note that, by (f), the $R_Z(\theta)$ gate commutes through the CZ unchanged. The rest of the analysis is identical to Example ??.

EXAMPLE 3.7 Now, suppose we chain together two teleportation gadgets, taking the output of one as the input ϕ to another. Then we are applying the following gate:

$$X^{m_2} HR_Z(\theta_2) X^{m_1} HR_Z(\theta_1) \stackrel{(\pi)}{=} X^{m_2} H X^{m_1} R_Z((-1)^{m_1} \theta_2) HR_Z(\theta_1) \quad (3.7)$$

$$\stackrel{(cc)}{=} X^{m_2} Z^{m_1} HR_Z((-1)^{m_1} \theta_2) HR_Z(\theta_1) \quad (3.8)$$

Chaining three gadgets, we are applying the following gate:

$$X^{m_3} HR_Z(\theta_3) X^{m_2} HR_Z(\theta_2) X^{m_1} HR_Z(\theta_1) \quad (3.9)$$

$$= X^{m_3} HR_Z(\theta_3) (X^{m_2} Z^{m_1} HR_Z((-1)^{m_1} \theta_2) X^{m_1} HR_Z(\theta_1)) \quad (3.10)$$

$$\stackrel{(\pi)}{\stackrel{(cc)}}{=} X^{m_3+m_1} Z^{m_2} (HR_Z((-1)^{m_2} \theta_3) HR_Z((-1)^{m_1} \theta_2) X^{m_1} HR_Z(\theta_1)). \quad (3.11)$$

Since $X^2 = Z^2 = I$, it is not too hard to see that chaining this gadget any number of times will result in an overall Pauli error of $X^m Z^n$ for some bits m, n .

The formal definition of the MBQC protocol incorporates these errors into the definition.

DEFINITION 3.8 We use the formal definition from [Bro+07], known as the **Measurement Calculus**. Denote an arbitrary single qubit state by

$$|+\theta,\phi\rangle := \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \quad (3.12)$$

and denote its orthogonal state by

$$|+\theta,\phi\rangle := \cos\left(\frac{\theta}{2}\right)|0\rangle - e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle \quad (3.13)$$

where $\theta \in [0, \pi]$ and $\phi \in [0, 2\pi)$. A computation in this model is a series of the following deterministic **commands**:

1. Preparation: 1-qubit preparations N_i , to prepare qubit i in the state $|+\rangle$;
2. Entanglement: 2-qubit entangling operators $E_{i,j} := Z_{i,j}$;
3. Measurement: 1-qubit measurements $M_i^{\lambda,\alpha}$ on planes $\lambda \in \{(X,Y), (X,Z), (Y,Z)\}$, defined as projections into

- (i) $|\pm_{(X,Y),\alpha}\rangle := |+\pi/2,\alpha\rangle$ for $\lambda = (X,Y)$,
- (ii) $|\pm_{(X,Z),\alpha}\rangle := |+\alpha,0\rangle$ for $\lambda = (X,Z)$, and
- (iii) $|\pm_{(Y,Z),\alpha}\rangle := |+\alpha,\pi/2\rangle$ for $\lambda = (Y,Z)$,

with the convention that $|+\theta,\phi\rangle\langle+\theta,\phi|$ corresponds to the outcome 0 and $|-\theta,\phi\rangle\langle-\theta,\phi|$ corresponds to 1; and

4. Correction: 1-qubit Pauli corrections X_i and Z_i .

The measurements $M_i^{\lambda,\alpha}$ may depend on **signals**, namely sums of some outcomes of previous measurements mod 2. These represent the Pauli errors of the form $X^m Z^n$ being carried through, for some bits m, n . A computation can then be expressed as a series of preparation, entanglement, measurement, and correction commands, strictly in this order, satisfying the following conditions:

- (a) No command depends on an outcome not yet measured;
- (b) No command acts on a qubit already measured;
- (c) No command acts on a qubit not yet prepared, unless it is an input qubit; and
- (d) A qubit i is measured if and only if i is not an output.

Equivalently, a computation may be defined as a **measurement pattern** on a some graph state $|G\rangle$. A measurement pattern $\mathcal{B} = (V, I, O, S)$ consists of

- a finite set of qubits $V \subset V(G)$,

- an input set $I \subset V(G)$, possibly overlapping with
- an output set $O \subset V(G)$,
- and a finite sequence of measurement and correction commands S .

that satisfy the above conditions (a) to (d).

Note that the measurement bases available in the Measurement Calculus model are different from the measurements in Examples 3.5 and 3.6. Thus, it will be convenient to find the operations applied by measurements in the XY, YZ, and XZ planes.

LEMMA 3.9 Let $|\psi\rangle = a|0\rangle + b|1\rangle$ be an arbitrary one-qubit input state. Prepare a second qubit $|+\rangle$, apply CZ, and measure the first qubit in a state

$$|m\rangle = u|0\rangle + v|1\rangle. \quad (3.14)$$

Conditioned on this measurement outcome, the unnormalized output state on the second qubit is

$$H \begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix} |\psi\rangle. \quad (3.15)$$

In particular, for the standard measurement planes we obtain:

Plane	Measurement state	ZX effect	Unnormalized output	Unitary?
XY	$\frac{1}{\sqrt{2}}(0\rangle + e^{-i\alpha} 1\rangle)$	$\textcircled{\alpha}$ —	$\frac{1}{\sqrt{2}}HP(\alpha) \psi\rangle$	Always
YZ	$c 0\rangle + is 1\rangle$	$\textcircled{\alpha}$ —	$H \begin{pmatrix} c & 0 \\ 0 & -is \end{pmatrix} \psi\rangle$	Only if $ c = s $
XZ	$c 0\rangle + s 1\rangle$	$\textcircled{\alpha}$ — $\textcircled{\frac{\pi}{2}}$ —	$H \begin{pmatrix} c & 0 \\ 0 & s \end{pmatrix} \psi\rangle$	Only if $ c = s $

where $P(\alpha) = \text{Diag}(1, e^{i\alpha}) = \llbracket \textcircled{\alpha} \rrbracket$, $c = \cos(\frac{\alpha}{2})$, and $s = \sin(\frac{\alpha}{2})$.

Proof. Write $|\psi\rangle = a|0\rangle + b|1\rangle$. Prepare a second qubit $|+\rangle$ and apply CZ:

$$CZ(|\psi\rangle|+\rangle) = CZ((a|0\rangle + b|1\rangle)|+\rangle) = a|0\rangle|+\rangle + b|1\rangle|-\rangle. \quad (3.16)$$

Now project the first qubit onto $|m\rangle = u|0\rangle + v|1\rangle$. The resulting unnormalized state is

$$(\langle m| \otimes I)CZ(|\psi\rangle|+\rangle) = (\langle m| \otimes I)(a|0\rangle|+\rangle + b|1\rangle|-\rangle) \quad (3.17)$$

$$= a\langle m|0\rangle|+\rangle + b\langle m|1\rangle|-\rangle \quad (3.18)$$

$$= a\bar{u}|+\rangle + b\bar{v}|-\rangle \quad (3.19)$$

$$= H \begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix} |\psi\rangle. \quad (3.20)$$

For the XY -plane state, $|m\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{-i\alpha}|1\rangle)$,. So $\bar{u} = \frac{1}{\sqrt{2}}$, and $\bar{v} = \frac{e^{i\alpha}}{\sqrt{2}}$.. Hence

$$H \begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix} |\psi\rangle = H \begin{pmatrix} 1/\sqrt{2} & 0 \\ 0 & e^{i\alpha}/\sqrt{2} \end{pmatrix} |\psi\rangle = \frac{1}{\sqrt{2}} H \begin{pmatrix} 1 & 0 \\ 0 & e^{i\alpha} \end{pmatrix} |\psi\rangle = \frac{1}{\sqrt{2}} H P(\alpha) |\psi\rangle. \quad (3.21)$$

For the YZ -plane state, $|m\rangle = c|0\rangle + is|1\rangle$,. Then $\bar{u} = c$, and $\bar{v} = -is$.. Therefore

$$H \begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix} |\psi\rangle = H \begin{pmatrix} c & 0 \\ 0 & -is \end{pmatrix} |\psi\rangle. \quad (3.22)$$

For the XZ -plane state, $|m\rangle = c|0\rangle + s|1\rangle$,. Hence $\bar{u} = c$, and $\bar{v} = s$. This gives

$$H \begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix} |\psi\rangle = H \begin{pmatrix} c & 0 \\ 0 & s \end{pmatrix} |\psi\rangle. \quad (3.23)$$

Finally, since H is unitary, $H \begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix}$ is proportional to a unitary iff $\begin{pmatrix} \bar{u} & 0 \\ 0 & \bar{v} \end{pmatrix}$ is proportional to a unitary. A diagonal matrix $D = \begin{pmatrix} r_0 & 0 \\ 0 & r_1 \end{pmatrix}$ is proportional to a unitary iff

$$D^\dagger D = \begin{pmatrix} |r_0|^2 & 0 \\ 0 & |r_1|^2 \end{pmatrix} = \lambda I \quad (3.24)$$

for some $\lambda > 0$, which holds exactly when $|r_0| = |r_1|$. Thus the XY case is always unitary, while the YZ and XZ cases are unitary only when $|c| = |s|$. $\square$

LEMMA 3.10 The Pauli eigenstates occur as special cases of XY , XZ , and YZ plane measurements:

Pauli effect	Three-plane form	State	ZX effect
$\langle +_X $	$\langle +_{XY,0} $	$ +\rangle = \frac{1}{\sqrt{2}}(0\rangle + 1\rangle)$	$\bigcirc -$
$\langle -_X $	$\langle +_{XY,\pi} $	$ -\rangle = \frac{1}{\sqrt{2}}(0\rangle - 1\rangle)$	$\bigcirc(\pi) -$
$\langle +_Y $	$\langle +_{XY,\pi/2} $	$ -i\rangle := \frac{1}{\sqrt{2}}(0\rangle - i 1\rangle)$	$\bigcirc(-\frac{\pi}{2}) -$
$\langle -_Y $	$\langle +_{XY,-\pi/2} $	$ i\rangle := \frac{1}{\sqrt{2}}(0\rangle + i 1\rangle)$	$\bigcirc(\frac{\pi}{2}) -$
$\langle +_Z $	$\langle +_{XZ,0} = \langle +_{YZ,0} $	$ 0\rangle$	$\bullet -$
$\langle -_Z $	$\langle +_{XZ,\pi} = \langle +_{YZ,\pi} $	$ 1\rangle$	$\bullet(\pi) -$

Consequently, X - and Y -basis measurements are contained in the XY -plane model, while Z -basis measurements require the XZ (or YZ) plane,

Proof. The XY -plane cases follow directly from substituting the relevant angles:

$$|+_{XY,0}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i0} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = |+_X\rangle, \quad (3.25)$$

$$|+_{XY,\pi}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = |-_X\rangle. \quad (3.26)$$

Similarly,

$$|+_{XY,\pi/2}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{i\pi/2} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle + i |1\rangle) = |+_Y\rangle, \quad (3.27)$$

$$|+_{XY,-\pi/2}\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{-i\pi/2} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle - i |1\rangle) = |-_Y\rangle. \quad (3.28)$$

For the Z -basis states, use the XZ plane:

$$|+_{XZ,0}\rangle = \cos(0) |0\rangle + \sin(0) |1\rangle = |0\rangle = |+_Z\rangle, \quad (3.29)$$

$$|+_{XZ,\pi}\rangle = \cos \frac{\pi}{2} |0\rangle + \sin \frac{\pi}{2} |1\rangle = |1\rangle = |-_Z\rangle. \quad (3.30)$$

Finally, every XY -plane state has the form

$$|+_{XY,\alpha}\rangle = \frac{1}{\sqrt{2}} |0\rangle + \frac{e^{i\alpha}}{\sqrt{2}} |1\rangle. \quad (3.31)$$

Hence both computational-basis amplitudes have magnitude $1/\sqrt{2}$. Therefore no XY -plane state can equal $|0\rangle$ or $|1\rangle$, even up to global phase. Thus Z -basis measurements are not contained in the XY -plane measurement model. $\square$

In this thesis, we are interested in whether there exist measurement patterns on comparability grids that result in universal quantum computation. It remains to formally define what it means to use comparability grids as a resource for universal quantum computation.

3.3 Resource States

DEFINITION 3.11 A **resource state** is a state, typically containing entanglement, on which a measurement pattern can be applied to conduct measurement-based quantum computation. Under the Measurement Calculus (Definition 3.8), a resource state is a graph state.¹⁰

DEFINITION 3.12 A **family** of resource states is some collection of resource states. A family of resource states Ψ is **universal for MBQC** if, for every n -qubit state $|\phi\rangle$, there exists some resource state $|\Psi_n\rangle \in \Psi$ and a measurement pattern $\mathcal{B}$ such that the output state is $|\phi\rangle$. We say a family of resource states is **n -qubit universal for MBQC** if this holds for all m -qubit $|\phi\rangle$ with $m \leq n$. When the meaning is clear, we may drop the phrase “for MBQC” and simply say a family of resource states is universal.

¹⁰We will see in the next subsection how results of the universality of graph states may extend to the universality of non-graph states.

We start by reciting some well-known results related to graph states.

FACT 3.13 There is a unique quantum circuit construction for every graph state, up to reordering of C-Z.

Proof. By (f), CZ gates commute past each other. Then each pair of qubits (vertices) has zero or one CZs (edges) between them. $\square$

FACT 3.14 Let G be a graph with maximum degree Δ . Then $|G\rangle$ can be constructed in depth $\Delta + 1$.

Proof. Vizing's theorem states that any graph with maximum degree Δ admits an edge coloring with $\Delta + 1$ colors such that no two edges that share a vertex have the same color. All edges of the same color can be implemented in the graph state in the same layer of CZs. Thus, at most $\Delta + 1$ layers are needed. $\square$

FACT 3.15 Given a graph state $|G\rangle$, a measurement in the YZ plane is equivalent to vertex deletion.

Proof. Consider a graph state $|G\rangle$ in its ZX-Calculus representation.

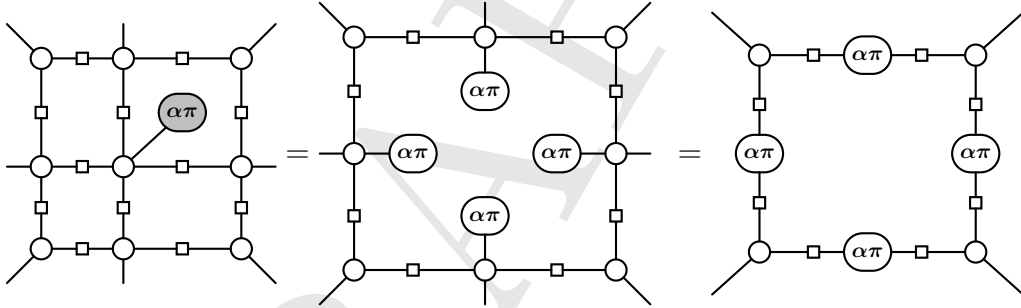


Figure 2: YZ plane measurements correspond to vertex deletion.

By Lemma 3.9, the ZX effect of a YZ plane measurement is a $\textcircled{a\pi}$ where a is the measurement outcome bit. By (cp), we may push the $\textcircled{a\pi}$ through the vertex being measured, deleting it in the process. The result is a graph state with the measured vertex removed, and an additional phase of $a\pi$ on all vertices connected to the measured vertex. $\square$

FACT 3.16 Given a graph state $|G\rangle$, applying $\sqrt{X} = -\textcircled{\pi/2}$ on a vertex v and $\sqrt{Z} = -\textcircled{\pi/2}$ on $N_G(v)$ is equivalent to a local complementation around v , and yields $|G \star v\rangle$. Applying a Y plane measurement on v instead gives $|G \star v - v\rangle$ with additional phases.

Proof. The first claim is Lemma 2.29 and the second is Lemma 2.30. $\square$

FACT 3.17 Given a graph state $|G\rangle$, applying H on vertices u, v and Z on $N_G(u) \cap N_G(v)$ gives $|G \wedge uv\rangle$. Applying a X plane measurement on both u and v instead gives $|G \wedge uv - u - v\rangle$ with additional phases.

Proof. The first claim is Lemma 2.32 and the second is Lemma 2.33. $\square$

DEFINITION 3.18 A $m \times n$ grid graph $G = (V, E)$ has

$$V = \{(i, j) \mid i \in [m], j \in [n]\}, \quad (3.32)$$

$$E = \{\{(i_1, j_1), (i_2, j_2)\} \mid (i_1, j_1), (i_2, j_2) \in V, |i_2 - i_1| + |j_2 - j_1| = 1\} \} \quad (3.33)$$

In other words, m rows and n columns with edges between nodes of taxicab distance 1. Denote such graphs as $\text{Grid}_{m \times n}$. Let Grid denote the family of all such graphs.

THEOREM 3.19 The Grid is universal.

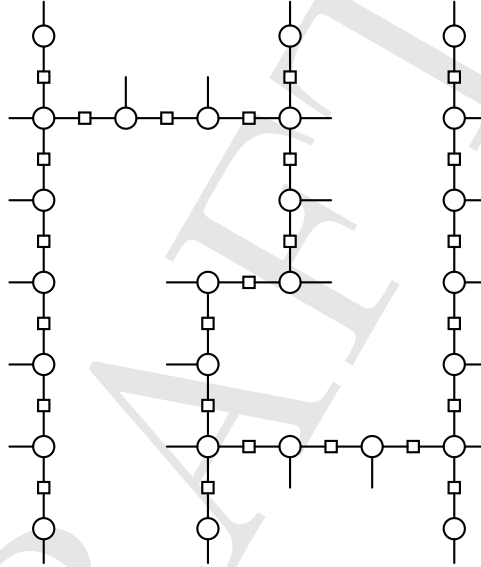


Figure 3: A Grid graph state with some vertices removed, creating three vertical “wires” for a three qubit computation in the MBQC model. The horizontal connections between wires may entangle the qubits, given the right measurements.

Proof. The full proof is quite involved and very tedious. The original proof is given in [RB01], and a more accessible proof is available in [RBB03]. A modern proof through the ZX-Calculus is available in Chapter 9 of [KW24].

There are three great difficulties in the proof. We informally discuss them. First, we need to be able to implement arbitrary single qubit gates. This is done by deleting (via Z plane measurements) vertices to create line graphs, each resulting in a chain like Example 3.6. It can be shown that this allows the application of arbitrary single qubit gates up to some Pauli error.

Second, we need to be able to entangle qubits. If the two line graphs containing the qubits we wish to entangle are adjacent, i.e. there are no other line graphs between them,

then creating a orthogonal line of vertices between the wires may entangle them. Now, if they are not adjacent, we can swap qubits between lines (by implementing CNOT thrice) until they become adjacent.

Third, we need to ensure measurement is done in the correct order. For example, suppose we delete all vertices connecting the input and output vertices. Then this computation is deterministic and not invertible, and thus does not implement a unitary. As it turns out, there is a criterion called *flow* that ensures this does not happen, as well as variants of flow such as *gflow* and *Pauli flow* that further ensure that we can deal with the Pauli errors. Interested readers should see [MB23; Bac+21; KW24] for more details. $\square$

3.4 Universality Criteria

Given the difficulty of the proof of Theorem 3.19, we now turn our attention to the development of other techniques to determine the universality of a family of resource states. We start by recalling some useful facts regarding regular quantum circuit universality.

FACT 3.20 Any 1-qubit universal gate set and a 2 qubit entangling gate is universal [BB01].

Proof. The proof in [BB01] turns out to be quite technical, involving many tricks from Lie Algebra. It surmounts to proving the following restatement of the theorem: Let V be a 2-qubit gate. The following are equivalent:

1. The collection of all 1-qubit gates $\mathcal{A}$ together with V is QQ-universal.
2. The collection of all 1-qubit gates $\mathcal{A}$ together with V is exactly QQ-universal.
3. V is not primitive, or **imprimitive**.

Here, say V is **primitive** iff $V = S \otimes T$ or $V = (S \otimes T)P$ for some 1-qubit gates S and T , with P being the swap gate. $\square$

THEOREM 3.21 Any universal gate set must include at least one multi-qubit entangling gate, as it is impossible to entangle using only single-qubit gates.

Proof. This follows from Fact 3.20. $\square$

THEOREM 3.22 Let **Line** denote the family of graphs made out of line graphs. Then **Line** is not a universal resource.

Proof. This follows from Theorem 3.21. $\square$

We now turn to universality criteria specifically for MBQC. We start by addressing why we only consider CQ-universality rather than QQ-universality.

THEOREM 3.23 This is Observation 2 of [Nes+07]. Every CQ-universal resource can be made QQ-universal by allowing quantum inputs to be coupled into the resource state using the same coupling method, i.e. CZ gates.

DEFINITION 3.24 Let **LO** denote local operations, i.e. any single qubit unitary and measurement. Let **LC** denote local Clifford operations, i.e. any ZX diagram with spiders of arity at most 2 and phases multiples of $\frac{\pi}{2}$. Let **CC** denote **classical communication**, which represents the measurement correction and Pauli error tracking done on-the-fly in MBQC.

States $|\psi\rangle$ and $|\phi\rangle$ both on n qubits are called **LU-equivalent** if there is a unitary $U = \otimes_{i=1}^n U_i$, where every U_i is a local unitary, such that $|\psi\rangle = U|\phi\rangle$. In this case, write $|\psi\rangle =_{\text{LU}} |\phi\rangle$. Now, if every U_i is made up of only local Clifford gates, say they are **LC-equivalent** and write $|\psi\rangle =_{\text{LC}} |\phi\rangle$.

THEOREM 3.25 This is the main result of [VDD04]. Graph states $|G_1\rangle$ and $|G_2\rangle$ are **LC-equivalent** iff there is a series of local complementations that transform one graph to another.

THEOREM 3.26 This is the main result of [Ji+08]. There are graphs that are **LU-equivalent** but not **LC-equivalent**. On the other hand, there are some types of graphs which are **LU-equivalent** iff they are **LC-equivalent** [GN07; VDD05].

THEOREM 3.27 A family of states Φ is a universal resource for MBQC if for all n -qubit states $|\psi\rangle$, $\exists |\phi\rangle \in \Phi$ with $m \geq n$ qubits, such that the transformation $|\phi\rangle \rightarrow |\psi\rangle +^{m-n}$ is possible deterministically with **LO** and **CC**. We denote this by $|\phi\rangle \geq_{\text{LOCC}} |\psi\rangle$.

Proof. This follows almost immediately from the definition. Suppose Φ is a universal resource. Then the above is certainly true, since a measurement pattern is made up of only **LO** and **CC**. Now suppose the above holds. It is clear that the resource state is universal. Thus the above is a sufficient and necessary condition. $\square$

THEOREM 3.28 [From [Van+06]] Let Ψ be a family of resource states and $|\phi\rangle, |\phi'\rangle$ arbitrary states with the same number of qubits. Let $E : |\phi\rangle \rightarrow \mathbb{R}$ be a function that satisfies the following properties:

1. $E(|\phi\rangle) \geq E(|\phi'\rangle)$ whenever $|\phi\rangle \geq_{\text{LOCC}} |\phi'\rangle$;
2. $\sup_{|\phi\rangle} E(|\phi\rangle) > \sup_{|\psi\rangle \in \Psi} E(|\psi\rangle)$.

Then Ψ cannot be a universal resource.

Proof. The first condition states that **LOCC** cannot increase this measure, while the second says that the supremum value of E , which can be ∞ , is not reached among all $|\psi\rangle \in \Psi$. If both were true, then clearly universality is not possible. $\square$

The above theorem is very general and consequently very powerful, as we will soon see. However, note that satisfying the converse of the second condition for some measure E is not sufficient to guarantee universality. For example, the family of *toric code states* have unbounded entanglement width but are not universal [BR07].

THEOREM 3.29 Consider the family of all star graphs **Star**. A **star graph** on n vertices is a tree where one vertex has degree $n - 1$ and all other vertices have degree 1. Then **Star** is not a universal resource.

We choose to prove a more powerful result.

LEMMA 3.30 Every tree has rank width at most 1.

Proof. Let $G = (V, E)$ be a tree. We will construct a rank decomposition where every cut has cut-rank at most 1. Root the tree G at some vertex r . For each vertex $v \neq r$, let G_v denote the subtree consisting of v and all of its descendants.

Let $A_v := V(G_v)$. Since G is a tree, there is exactly one edge between A_v and $V \setminus A_v$, namely the edge from v to its parent. Therefore, the cut matrix M_{A_v} has exactly one nonzero entry. Hence

$$\rho_G(A_v) = \text{rank}_{\mathbb{F}_2} M_{A_v} \leq 1.$$

Now choose a binary decomposition tree T that recursively separates rooted subtrees of G . Every cut appearing in this decomposition is of the form $A_v \mid V \setminus A_v$ or a union of such subtrees attached to a common vertex. In either case, because G is a tree, the edges crossing the cut are incident to a single vertex on one side of the cut. Thus the corresponding bipartite adjacency matrix has all nonzero entries contained in a single row or a single column, so its rank over $\mathbb{F}_2$ is at most 1. Hence this rank decomposition has width at most 1. Therefore $\text{rwd}(G) \leq 1$. $\square$

DEFINITION 3.31 Let Φ and Ψ be families of resource states. We say that Φ and Ψ are **LU-equivalent**, and write $\Phi =_{\text{LU}} \Psi$, if the following holds:

1. For every $|\phi\rangle \in \Phi$, there exists $|\psi\rangle \in \Psi$ such that $|\phi\rangle =_{\text{LU}} |\psi\rangle$, and
2. for every $|\psi\rangle \in \Psi$, there exists $|\phi\rangle \in \Phi$ such that $|\psi\rangle =_{\text{LU}} |\phi\rangle$.

Define **LC-equivalent** correspondingly.

LEMMA 3.32 Let Φ and Ψ be LU-equivalent families of resource states. Then Φ is a universal resource for MBQC if and only if Ψ is a universal resource for MBQC.

Proof. Suppose Φ is universal, and let $|\rho\rangle$ be any n -qubit target state. By universality, there exists $|\phi\rangle \in \Phi$ on $m \geq n$ qubits such that $|\phi\rangle \geq_{\text{LOCC}} |\rho\rangle$. Since $\Phi =_{\text{LU}} \Psi$, there exists $|\psi\rangle \in \Psi$ such that $|\psi\rangle =_{\text{LU}} |\phi\rangle$. Local unitaries are reversible LOCC operations, so $|\psi\rangle \geq_{\text{LOCC}} |\phi\rangle$. Composing LOCC transformations gives $|\psi\rangle \geq_{\text{LOCC}} |\rho\rangle$. Thus Ψ is universal. The reverse implication is identical. $\square$

4 The Comparability Grid

4.1 Circle Graphs

DEFINITION 4.1 The $m \times n$ **comparability grid** is a graph $G = (V, E)$ with

$$V = \{(i, j) \mid i \in [m], j \in [n]\} \quad (4.1)$$

$$E = \{\{(i_1, j_1), (i_2, j_2)\} \mid (i_1, j_1), (i_2, j_2) \in V, i_2 \geq i_1, j_2 \geq j_1\}\} \quad (4.2)$$

In other words, m rows and n columns with edges defined by the partial order $\leq$. Denote this graph as $\text{CGrid}_{m \times n}$. Let CGrid denote the family of all such graphs.

EXAMPLE 4.2 Figure ?? shows the 3×3 comparability grid:

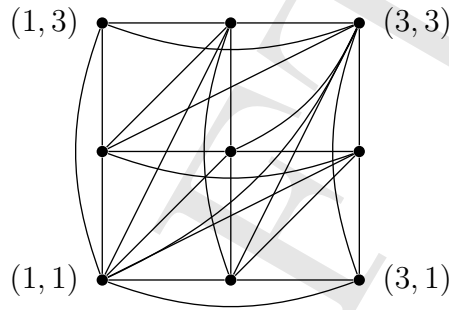


Figure 4: $\text{CGrid}_{3 \times 3}$.

4.2 Subgraphs of the Comparability Grid

One idea it to ask how close “vertex deletion” can get us to universal quantum computation. Namely, can we delete vertices to get a **Grid**? If not, how close can we get?

DEFINITION 4.3 Define a subset of a CGrid state as the graph corresponding to a state in CGrid with some vertices measured in the Z basis. Graph theoretically, a subset of a CGrid is a CGrid with some vertices removed (alongside any edges connected to said removed vertices).

FACT 4.4 Any subset H with n vertices of any CGrid can be uniquely identified as a permutation of $[n]$, denoted $\pi \in S_n$. Denote this by $\pi(H)$. Similarly, let $H(\pi)$ denote the subgraph corresponding to π .

Proof. This proof only works for finite subsets, but there is a need to define it for infinite-sized vertex subsets. Consider an arbitrary subset $H = (V, E)$ with $|V| = n$ of a fixed CGrid G . We claim this subgraph is isomorphic to the graph $H' = (V', E')$, where

1. $V' = \{(x + y/n, y + x/n) \mid (x, y) \in V\}$
2. $E' = \{\{(x_1 + y_1/n, y_1 + x_1/n), (x_2 + y_2/n, y_2 + x_2/n)\} \mid \{(x_1, y_1), (x_2, y_2)\} \in E\}$.

That is, H' is the graph whose x' and y' values have been shifted rightwards and upwards, respectively. Since $|V| = n$, there is a clear bijection from V to V' . The bijection between edges follows as $|E| = |E'|$.

We claim the data of H' can be completely expressed as follows: Sort the vertices in V' by their x' -coordinate. No two distinct vertices can share an x' coordinate: We proceed by contradiction. If $x_1 = x_2$

$$x_1 + \frac{y_1}{n} = x_2 + \frac{y_2}{n} \implies y_1 = y_2 \quad (4.3)$$

but then they are not distinct vertices. If $x_1 \neq x_2$, without loss of generality let $x_1 > x_2$ so $x_1 = x_2 + c$, $c \in \mathbb{N}$. Then

$$x_1 + \frac{y_1}{n} = x_2 + \frac{y_2}{n} \implies c = \frac{y_1}{n} - \frac{y_2}{n} \implies c = \frac{1}{n}(y_1 - y_2) \quad (4.4)$$

Since $y_1, y_2 \leq n$, the RHS ≤ 1 . Contradiction. No two distinct vertices share a y' coordinate either, by a similar proof. Re-label each vertex by their order, from least to greatest, in both x' and y' . Edges still form by the partial ordering rule in Equation 4.2. Reading the x' coordinate as the index of a n -tuple and the y' coordinate as the values, we obtain a permutation of n as each x' , y' is distinct. $\square$

FACT 4.5 Two subsets H, H' of a CGrid may have $\pi(H) \neq \pi(H')$ yet $H = H'$ up to vertex relabeling.

Proof. Take subgraphs with vertices $V = \{(1, 1), (1, 2)\}$ and $V' = \{(1, 1), (2, 1)\}$ and their induced edges. The first has permutation $(1, 2)$ and the second $(2, 1)$ but the graphs are clearly isomorphic. $\square$

FACT 4.6 Let $\pi \in S_n$ and define its reverse-complement

$$\pi^{\text{rc}}(i) := (n + 1) - \pi(n + 1 - i). \quad (4.5)$$

Then the induced CGrid subgraphs satisfy $H(\pi) \cong H(\pi^{\text{rc}})$.

Proof. Write vertices of $H(\pi)$ as $v_i = (i, \pi(i))$. Define a bijection

$$f : v_i \mapsto v'_{n+1-i} := (n + 1 - i, n + 1 - \pi(i)). \quad (4.6)$$

The image vertex set is exactly $\{(k, \pi^{\text{rc}}(k)) \mid k \in [n]\}$, so f is a bijection from $V(H(\pi))$ to $V(H(\pi^{\text{rc}}))$. An edge in a comparability grid exists between v_i and v_j iff

$$(i - j)(\pi(i) - \pi(j)) > 0, \quad (4.7)$$

equivalently $(i < j \ \& \ \pi(i) < \pi(j))$ or $(i > j \ \& \ \pi(i) > \pi(j))$. But under f we have

$$(n + 1 - i) - (n + 1 - j) = -(i - j), \quad (4.8)$$

$$(n + 1 - \pi(i)) - (n + 1 - \pi(j)) = -(\pi(i) - \pi(j)), \quad (4.9)$$

so the product $(i - j)(\pi(i) - \pi(j))$ is preserved. Hence $\{v_i, v_j\}$ is an edge iff $\{f(v_i), f(v_j)\}$ is an edge. Therefore f is a graph isomorphism. $\square$

THEOREM 4.7 Let $\pi(H)$ denote a permutation of n capturing the data of a finite subset $H = (V, E) \leq G$ with $|V| = n$, for some $G \in \text{CGrid}$. The degree sequence of $\pi(H)$ can be constructed algorithmically in $O(n \log n)$ time and $O(n)$ space.

Proof. For any $\pi \in S_n$, let an **inversion** be a pair (i, j) , $i \neq j \in [n]$, such that

$$i < j \text{ and } \pi[i] > \pi[j]. \quad (4.10)$$

(i, j) is a **non-inversion** if

$$i < j \text{ and } \pi[i] < \pi[j]. \quad (4.11)$$

Since $i \neq j$, every (i, j) is either an inversion or non-inversion. Thus

$$\#\text{non-inversions} = \binom{n}{2} - \#\text{inversions}. \quad (4.12)$$

At position $j \in [n]$ of the permutation,

$$|\{i < j \mid \pi[i] < \pi[j]\}| = \text{prefixSum}(\pi[j] - 1). \quad (4.13)$$

Here, prefix sum refers to the number of indices with index $\leq j$ and value $\leq \pi[j]$. The -1 makes $\leq$ become $<$. In other words, for each j , we count the number of indices $i < j$ such that $\pi[i] < \pi[j]$.

It is well-known that a prefix sum can be computed in $\Theta(\log n)$ time using a binary index tree or Fenwick tree. Since there are n iterations, the total time is $\Theta(n \log n)$. The Fenwick tree is the only allocation, and only needs $\Theta(n)$ more space.

Run the above procedure on the reverse of $\pi(H)$. This counts, for each j , the number of indices $i < j$ such that $\pi[i] > \pi[j]$. Adding the two values for each index gives the degree of the corresponding vertex. Denote this algorithm as **double inversion**. $\square$

We use double-inversion rather than directly constructing the edges due to the following fact:

FACT 4.8 In a subgraph H with n vertices, there are at most $O(n^2)$ edges. Thus, enumerating the edges of some $\pi(H)$ cannot be better than $O(n^2)$ in the worst general case.

THEOREM 4.9 It is sufficient to explore subgraphs with at most $O(n)$ edges. Thus, enumerating the edges of some $\pi(H)$ cannot be worse than $O(n)$ in the worst general case.

Proof. **This needs some formalization.** The idea is to use early termination. Suppose we identify, even in the first pass of Theorem 4.7, that the current graph has “too many” edges or “too many” vertices with “high degree” to be the construction with optimal minimum number of edges compared to the **Grid**. Since every iteration of the algorithm through the vertices in both passes can only add 0 or more edges, we can early terminate the entire branch. We call these early terminations **termination strategies**. $\square$

THEOREM 4.10 When enumerating every possible subgraph $H = (V, E)$ of some CGrid with $|V| = n$, it is more than sufficient to enumerate every possible permutation of $[n]$.

Proof. Clearly, there is a map $f : \{\forall H\} \rightarrow S_n$. If there is a subset that does not have a representation in S_n , this would violate Fact 4.4. This is more than sufficient as Fact 4.5 shows the permutation representation is not unique. $\square$

Before we enumerate the termination strategies we will use, it may be helpful to define some notation.

DEFINITION 4.11 Given any Grid, say a vertex is a corner vertex if it has degree 2 in the grid. Say it is an edge if it has degree 3, and inner if it has degree 4. In the context of a subgraph H of some CGrid, say a vertex in H is a **potential corner** if it has degree at least 2. Similarly for **potential edge** and **potential inner**.

Furthermore, say a vertex in H is **invalid** if it has degree less than 2. We say invalid as we wish to find the number of maximum additional edges, not minimum number of missing edges. In other words, if H has an invalid vertex, no Grid G with $|V(H)| = |V(G)|$ is a subgraph of H .

FACT 4.12 To enumerate all possible permutations of $[n]$, it suffices to consider all possible choices for the first entry, then all possible remaining choices for the second entry for each chosen first entry, and henceforth. This task is massively parallelizable by seeding $k \leq n$ first entries.

[This needs to be made more formal.](#) Thus we can say we **terminate** a set of permutations in S_n if the processing of $j \leq n$ entries in either pass of Theorem 4.7 triggers a termination strategy.

DEFINITION 4.13 [This needs to be edited for brevity.](#) We now enumerate the various termination strategies. Throughout, let $H = (V, E)$ be the subgraph of some CGrid, and let $\text{Grid}_{k \times n}$ be the grid we are comparing H with. Let first and second pass refer to the two passes of Theorem 4.7.

Suppose we set a bound b on the maximum number of edges.

1. If in the second pass, a vertex is found invalid, terminate. No vertex can be found invalid in the first pass.
2. If the sum of degrees, divided by 2 due to the Handshaking lemma, is larger than b , terminate.
3. Suppose a vertex has degree $> b + 4$. Then this subgraph invalidates b . [This may be equivalent to the previous strategy.](#)
4. The distribution of degrees should be approximately constant. In other words, the additional edges b should be spread around the graph. This can be computed by keeping a counter of the number of degree $1, \dots, b + 4$ vertices found in the second pass.

THEOREM 4.14 All the termination strategies above can be done in $\Theta(1)$ time and $\Theta(1)$ space.

Proof. Most strategies can be done in such bounds by keeping a running sum, running a constant number of if statements, and other $\Theta(1)$ time and space operations. $\square$

Recall that our goal is to identify whether vertex deletion alone can get us to a subgraph isomorphic to some **Grid** graph. This is surprisingly true for $2 \times n$ grids for all $n \in \mathbb{N}$.

THEOREM 4.15 The $2 \times n$ **Grid** is obtainable from a sufficiently large **CGrid** through vertex deletion alone.

Proof. Fix $G = \text{CGrid}_{n \times n}$. Define the i th **anti-diagonal** as

$$A_n := \{(i, j) \mid i + j = n + 1\} \quad (4.14)$$

For example, $A_1 = \{(1, 1)\}$ and $A_2 = \{(1, 2), (2, 1)\}$. Delete all vertices not in some A_j or $A_{j+1} \setminus \{(j, 1), (1, j)\}$ for $j \in [n - 1]$. The remaining vertices and induced edges form a $2 \times n$ grid:

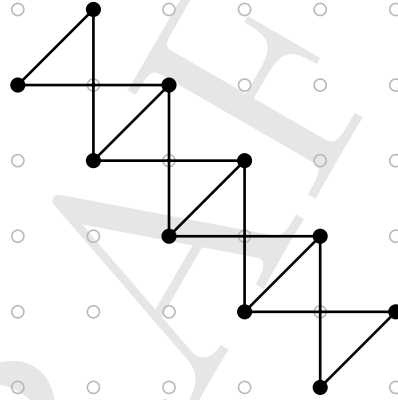


Figure 5: The 2×5 **Grid** subset in a $\text{CGrid}_{6 \times 6}$ using only vertex deletion.

To see that this is a grid, “twist” the above figure. $\square$

However, the above pattern does not generalize for larger widths.

FACT 4.16 It is not possible to get a $\text{Grid}_{3 \times n}$ subgraph from any size of **CGrid** using only vertex deletion.

Proof. Brute force using the algorithm described in Theorem 4.7. $\square$

That said, we can get quite close to a grid:

DEFINITION 4.17 Consider the following construction:

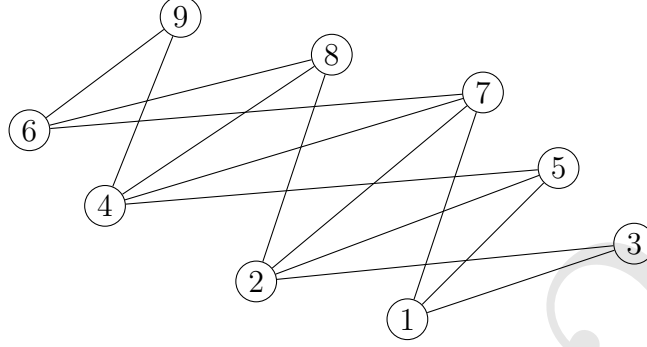


Figure 6: A subset close to a $\text{Grid}_{3 \times 3}$ from some large CGrid using only vertex deletion.

This forms the following subgrid:

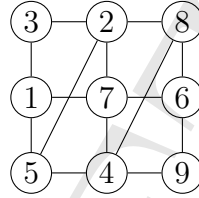


Figure 7: Tileable $\text{Grid}_{3 \times n}$ subgraph of CGrid with one additional edge every column.

This construction is tileable. That is, it generalizes to $\text{Grid}_{3 \times n}$ provided a sufficiently large CGrid .

Proof. I had an extremely complicated proof. The crux of the issue is that every time a new column is added, we need to ensure that it can "fit" in between (in either the x or y dimension) some previous existing points. Since our points need to be in $\mathbb{N} \times \mathbb{N}$, the simplest solution is to pre-allocate some gap between the nodes. However, this becomes extremely cumbersome very quickly.

An easier proof is to work graphically. That said, it would be great if the proof was in the ZX-representation of the CGrid , as this may highlight some important identities and tricks usable for even higher widths. $\square$

4.3 Vertex Minors

DEFINITION 4.18 A graph H is a vertex minor of a graph G if there is a sequence of vertex deletions and local complementations from G to H .

The above definition is a major motivation for our focus on vertex deletion in the previous section. Thus, it may make sense to swap the order of sections around

THEOREM 4.19 Deciding whether a graph H is a vertex-minor of G is **NP**-complete [DHW20].

DEFINITION 4.20 The definition of the **rank width** of a graph $G = (V, E)$ is quite involved and given in [OS06].

FACT 4.21 The rank width of a graph is preserved under local complementation. Vertex deletions may reduce rank-width, so the rank width of G is at least the rank width of any vertex-minor of G .

THEOREM 4.22 Every CGrid is a vertex-minor of a graph in every family of graphs with unbounded rank-width [cite:need]. Formally, there is a function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that every graph of rank-width $\geq f(n)$ has a vertex-minor isomorphic to $\text{CGrid}_{n \times n}$ [Gee+23].

THEOREM 4.23 Grid graphs contain all graphs as vertex minors. Formally, there is a function $f : G \rightarrow \mathbb{N}$ such a graph G is the vertex minor of $\text{Grid}_{f(G) \times f(G)}$.

Proof. [I would like to formalize the proof in Kunal's notes from James George Davies.](#) $\square$

THEOREM 4.24 If $\text{CGrid} \geq_{\text{LOCC}} \text{Grid}$ then unbounded rank-width is not sufficient for universality.

5 Algorithmic Approaches

5.1 Overview

5.2 Subgraph Algorithms

5.3 Gadget Extraction

In this section, we will focus on how we may extract a circuit gadget that captures the computational power of a tileable subgraph of a CGrid.

FACT 5.1 We start with a useful identity:

$$\begin{array}{c} q_0 \\ q_1 \\ q_2 \end{array} \begin{array}{c} \bullet \\ | \\ \oplus \end{array} = \begin{array}{c} q_0 \\ q_1 \\ q_2 \end{array} \begin{array}{c} \oplus \\ | \\ \oplus \end{array} \quad (5.1)$$

Proof. Note that

$$\begin{array}{c} \circ \\ | \\ \circ \end{array} \begin{array}{c} \bullet \\ | \\ \bullet \end{array} \stackrel{(b)}{=} \begin{array}{c} \bullet \\ | \\ \bullet \end{array} \begin{array}{c} \circ \\ | \\ \circ \end{array} \stackrel{(f)}{=} \begin{array}{c} \bullet \\ | \\ \bullet \end{array} \begin{array}{c} \circ \\ | \\ \circ \end{array} \stackrel{(f)}{=} \begin{array}{c} \bullet \\ | \\ \bullet \end{array} \begin{array}{c} \circ \\ | \\ \circ \end{array} \quad (5.2)$$

This completes the proof. $\square$

DEFINITION 5.2 We define a gadget that captures the computational power of the construction in Definition 4.17. Start with the standard MBQC protocol:

$$\begin{array}{c} \left\{ \begin{array}{c} |\psi\rangle \\ |+\rangle \\ |+\rangle \\ |+\rangle \end{array} \right\} \begin{array}{c} \bullet \\ | \\ \bullet \\ | \\ \bullet \\ | \\ \bullet \end{array} \begin{array}{c} R_z(\theta_1) \\ R_z(\theta_2) \\ R_z(\theta_3) \end{array} \begin{array}{c} H \\ H \\ H \end{array} \begin{array}{c} \nearrow \\ \nearrow \\ \nearrow \end{array} \end{array} \quad (5.3)$$

Pushing the gates through like in Equation 3.2:

(5.4)

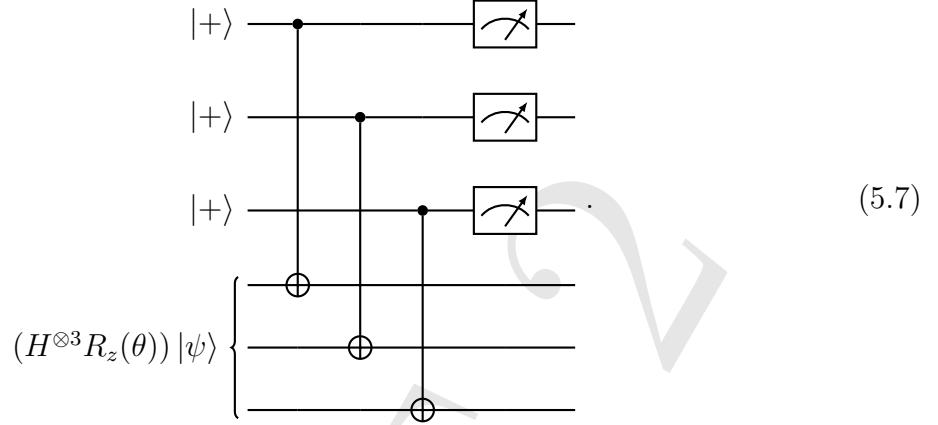
We can add the following CNOTs with affecting the circuit, like in Equation 3.3:

(5.5)

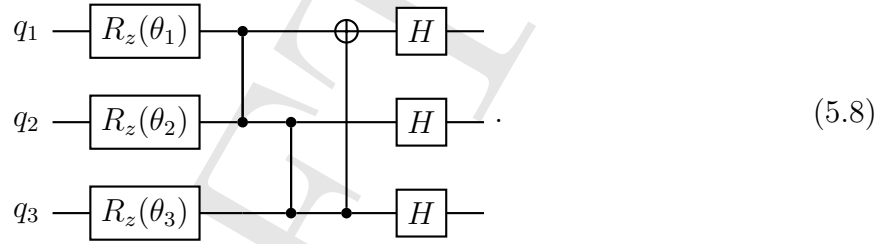
Applying the SWAP gate identity from Equation 3.4:

(5.6)

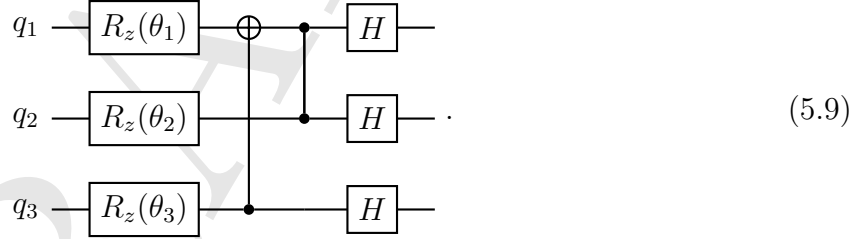
Finally, commuting the swap gates to the beginning gives



Dropping the Pauli gates, we get that we can apply blocks of



Call the above the **width-3 gadget**. Using Fact 5.1, the above gadget simplifies to



FACT 5.3 The group of Clifford operations, denoted $\langle C \rangle$, mod the group of Pauli operations, denoted $\langle P \rangle$, satisfies

$$|\langle C \rangle / \langle P \rangle| = 1,451,520 \quad (5.10)$$

We now introduce two new universality tests, one informal and one formal.

DEFINITION 5.4 (Informal universality tests). Replacing the tableau representation in the algorithm behind Fact ?? with general matrices, and increasing the number of angles, we should expect that the number of induced gates should increase dramatically. A proof should be given for this.

FACT 5.5 The previous algorithm does not indicate that the width-3 gadget is universal.

THEOREM 5.6 [SK17] provides an algorithm for determining the universality of a gate set, in terms of a linear algebra problem. Though it is not polynomial in terms of the number of qubits and gates, it is massively parallelizable. Ideally, I would like a GPU kernel that does much of this work.

5.4 Iterative Constructions

5.5 Lie Algebra Computations

DRAFT 2

6 Conclusions

6.1 Conjectures

I would like to produce a table of the minimum number of additional edges from attempting to subset $\text{Grid}_{k \times n}$ from CGrid . The cases $k = 2, 3$ already appear in the previous sections, but $k = 4, 5$ should be within reach in a few weeks. This is because of the following:

CONJECTURE 6.1 *This needs to be formalized:* Suppose we have a construction to subset a $k \times n$ subgraph H of some CGrid with a additional edges compared to $\text{Grid}_{k \times n}$. Suppose further than this construction produces b_k additional edges for every column added, and this b remains constant regardless of how many additional columns are added. Then a subgraph H' of any CGrid with additional edges compared to $\text{Grid}_{k+1 \times n}$ must have at least

$$b_{k+1} \geq \lceil b_k \frac{k+1}{k} \rceil \quad (6.1)$$

additional edges for every column added.

Proof. By contradiction and induction. The base case is $3 \times n$ where $b_3 = 1$, and $4 \times n$ where if $b_4 = 1$, then at least two rows have no additional edge. Deleting the one row connected to this additional edge and performing some transformations afterwards, which need to be formalized leaves a $3 \times n$ grid with fewer than b_3 additional edges per column. Contradiction. It is not quite clear whether this generalizes and hence is left as a conjecture. $\square$

CONJECTURE 6.2 Using the same notation as Conjecture 6.1, whatever construction H is minimal for k rows has H be a bipartite graph. In particular, it is of the form of two antidiagonal regions of the comparability grid. Since the bipartite graphs with n vertices are counted by C_n , the n th Catalan number, which is large but smaller than $n!$, significant early termination is still necessary to efficiently find the minimum number of additional edges for any k .

It is a fact that the number of additional edges must increase as k increases. In that sense, the $3 \times n$ case is the easiest case to check universality through algorithmic approaches for.

Here I want to provide a table of the results of the size of the groups generated using the width-3-gadget, and if there is enough compute, the width-4-gadget.

Another algorithmic direction is to include local complementation into the algorithmic search. However this is computationally quite expensive, because it means we cannot use the tuple subset representation of Fact 4.4. Furthermore, (local) graph isomorphism is now necessary to prevent checking the same graph multiple times, which further adds to the complexity of the algorithm.

6.2 Future Directions

Random measure idea.

References

- [09] *Mathematical Theory and Computational Practice: 5th Conference on Computability in Europe, CiE 2009, Heidelberg, Germany, July 19-24, 2009. Proceedings.* Springer Berlin Heidelberg, 2009. ISBN: 9783642030734. DOI: [10.1007/978-3-642-03073-4](https://doi.org/10.1007/978-3-642-03073-4). URL: <http://dx.doi.org/10.1007/978-3-642-03073-4>.
- [AC07] Samson Abramsky and Bob Coecke. *A categorical semantics of quantum protocols*. 2007. arXiv: [quant-ph/0402130](https://arxiv.org/abs/quant-ph/0402130) [quant-ph]. URL: <https://arxiv.org/abs/quant-ph/0402130>.
- [AG04] Scott Aaronson and Daniel Gottesman. “Improved simulation of stabilizer circuits”. In: *Physical Review A* 70.5 (Nov. 2004). ISSN: 1094-1622. DOI: [10.1103/PhysRevA.70.052328](https://doi.org/10.1103/PhysRevA.70.052328). URL: <http://dx.doi.org/10.1103/PhysRevA.70.052328>.
- [Aha+08] Dorit Aharonov et al. “Adiabatic Quantum Computation Is Equivalent to Standard Quantum Computation”. In: *SIAM Review* 50.4 (2008), pp. 755–787. ISSN: 00361445. URL: <http://www.jstor.org/stable/20454175> (visited on 01/27/2026).
- [Bac+21] Miriam Backens et al. “There and back again: A circuit extraction tale”. In: *Quantum* 5 (Mar. 2021), p. 421. ISSN: 2521-327X. DOI: [10.22331/q-2021-03-25-421](https://doi.org/10.22331/q-2021-03-25-421). URL: <http://dx.doi.org/10.22331/q-2021-03-25-421>.
- [Bar+95] Adriano Barenco et al. “Elementary gates for quantum computation”. In: *Phys. Rev. A* 52 (5 Nov. 1995), pp. 3457–3467. DOI: [10.1103/PhysRevA.52.3457](https://doi.org/10.1103/PhysRevA.52.3457). URL: <https://link.aps.org/doi/10.1103/PhysRevA.52.3457>.
- [BB01] Jean-Luc Brylinski and Ranee Brylinski. *Universal quantum gates*. 2001. arXiv: [quant-ph/0108062](https://arxiv.org/abs/quant-ph/0108062) [quant-ph]. URL: <https://arxiv.org/abs/quant-ph/0108062>.
- [BR07] Sergey Bravyi and Robert Raussendorf. “Measurement-based quantum computation with the toric code states”. In: *Phys. Rev. A* 76 (2 Aug. 2007), p. 022304. DOI: [10.1103/PhysRevA.76.022304](https://doi.org/10.1103/PhysRevA.76.022304). URL: <https://link.aps.org/doi/10.1103/PhysRevA.76.022304>.
- [Bro+07] Daniel E Browne et al. “Generalized flow and determinism in measurement-based quantum computation”. In: *New Journal of Physics* 9.8 (Aug. 2007), pp. 250–250. ISSN: 1367-2630. DOI: [10.1088/1367-2630/9/8/250](https://doi.org/10.1088/1367-2630/9/8/250). URL: <http://dx.doi.org/10.1088/1367-2630/9/8/250>.
- [Chi93] A. Chi-Chih Yao. “Quantum circuit complexity”. In: *Proceedings of 1993 IEEE 34th Annual Foundations of Computer Science*. 1993, pp. 352–361. DOI: [10.1109/SFCS.1993.366852](https://doi.org/10.1109/SFCS.1993.366852).
- [CK17] Bob Coecke and Aleks Kissinger. *Picturing Quantum Processes: A First Course in Quantum Theory and Diagrammatic Reasoning*. Cambridge University Press, Mar. 30, 2017. ISBN: 9781316219317. DOI: [10.1017/9781316219317](https://doi.org/10.1017/9781316219317). URL: <https://www.cambridge.org/core/books/picturing-quantum-processes/1119568B3101F3A685B>

- [Deu85] D. Deutsch. “Quantum Theory, the Church-Turing Principle and the Universal Quantum Computer”. In: *Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences* 400.1818 (1985), pp. 97–117. ISSN: 00804630. URL: <http://www.jstor.org/stable/2397601> (visited on 01/27/2026).
- [DHW20] Axel Dahlberg, Jonas Helsen, and Stephanie Wehner. “How to transform graph states using single-qubit operations: computational complexity and algorithms”. In: *Quantum Science and Technology* 5.4 (Sept. 2020), p. 045016. DOI: [10.1088/2058-9565/aba763](https://doi.org/10.1088/2058-9565/aba763). URL: <https://doi.org/10.1088/2058-9565/aba763>.
- [DP14] Ross Duncan and Simon Perdrix. “Pivoting makes the ZX-calculus complete for real stabilizers”. In: *Electronic Proceedings in Theoretical Computer Science* 171 (Dec. 2014), pp. 50–62. ISSN: 2075-2180. DOI: [10.4204/eptcs.171.5](https://doi.org/10.4204/eptcs.171.5). URL: <http://dx.doi.org/10.4204/EPTCS.171.5>.
- [Dun+20] Ross Duncan et al. “Graph-theoretic Simplification of Quantum Circuits with the ZX-calculus”. In: *Quantum* 4 (June 2020), p. 279. ISSN: 2521-327X. DOI: [10.22331/q-2020-06-04-279](https://doi.org/10.22331/q-2020-06-04-279). URL: <https://doi.org/10.22331/q-2020-06-04-279>.
- [Far+01] Edward Farhi et al. “A Quantum Adiabatic Evolution Algorithm Applied to Random Instances of an NP-Complete Problem”. In: *Science* 292.5516 (2001), pp. 472–475. DOI: [10.1126/science.1057726](https://doi.org/10.1126/science.1057726). eprint: <https://www.science.org/doi/pdf/10.1126/science.1057726>. URL: <https://www.science.org/doi/abs/10.1126/science.1057726>.
- [Gee+23] Jim Geelen et al. “The Grid Theorem for vertex-minors”. In: *Journal of Combinatorial Theory, Series B* 158 (2023). Robin Thomas 1962-2020, pp. 93–116. ISSN: 0095-8956. DOI: <https://doi.org/10.1016/j.jctb.2020.08.004>. URL: <https://www.sciencedirect.com/science/article/pii/S0095895620300794>.
- [GN07] D. Gross and M. Van den Nest. *The LU-LC conjecture, diagonal local operations and quadratic forms over GF(2)*. 2007. arXiv: [0707.4000](https://arxiv.org/abs/0707.4000) [quant-ph]. URL: <https://arxiv.org/abs/0707.4000>.
- [Got98] Daniel Gottesman. *The Heisenberg Representation of Quantum Computers*. 1998. arXiv: [quant-ph/9807006](https://arxiv.org/abs/quant-ph/9807006) [quant-ph]. URL: <https://arxiv.org/abs/quant-ph/9807006>.
- [Ji+08] Zhengfeng Ji et al. *The LU-LC conjecture is false*. 2008. arXiv: [0709.1266](https://arxiv.org/abs/0709.1266) [quant-ph]. URL: <https://arxiv.org/abs/0709.1266>.
- [JPV20] Emmanuel Jeandel, Simon Perdrix, and Renaud Vilmart. “Completeness of the ZX-Calculus”. In: *Logical Methods in Computer Science* Volume 16, Issue 2, 11 (June 2020). ISSN: 1860-5974. DOI: [10.23638/LMCS-16\(2:11\)2020](https://lmcs.episciences.org/5288). URL: <https://lmcs.episciences.org/5288>.
- [KW24] Aleks Kissinger and John van de Wetering. *Picturing Quantum Software*. Preprint, 2024.

- [MB23] Tommy McElvanney and Miriam Backens. “Flow-preserving ZX-calculus Rewrite Rules for Optimisation and Obfuscation”. In: *Electronic Proceedings in Theoretical Computer Science* 384 (Aug. 2023), pp. 203–219. ISSN: 2075-2180. DOI: [10.4204/eptcs.384.12](https://doi.org/10.4204/eptcs.384.12). URL: <http://dx.doi.org/10.4204/EPTCS.384.12>.
- [Nes+07] M Van den Nest et al. “Fundamentals of universality in one-way quantum computation”. In: *New Journal of Physics* 9.6 (June 2007), pp. 204–204. ISSN: 1367-2630. DOI: [10.1088/1367-2630/9/6/204](https://doi.org/10.1088/1367-2630/9/6/204). URL: <http://dx.doi.org/10.1088/1367-2630/9/6/204>.
- [OS06] Sang-il Oum and Paul Seymour. “Approximating clique-width and branch-width”. In: *Journal of Combinatorial Theory, Series B* 96.4 (2006), pp. 514–528. ISSN: 0095-8956. DOI: <https://doi.org/10.1016/j.jctb.2005.10.006>. URL: <https://www.sciencedirect.com/science/article/pii/S0095895605001528>.
- [Pre18] John Preskill. “Quantum Computing in the NISQ era and beyond”. In: *Quantum* 2 (Aug. 2018), p. 79. ISSN: 2521-327X. DOI: [10.22331/q-2018-08-06-79](https://doi.org/10.22331/q-2018-08-06-79). URL: <https://doi.org/10.22331/q-2018-08-06-79>.
- [RB01] Robert Raussendorf and Hans J. Briegel. “A One-Way Quantum Computer”. In: *Phys. Rev. Lett.* 86 (22 May 2001), pp. 5188–5191. DOI: [10.1103/PhysRevLett.86.5188](https://link.aps.org/doi/10.1103/PhysRevLett.86.5188). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.86.5188>.
- [RBB03] Robert Raussendorf, Daniel E. Browne, and Hans J. Briegel. “Measurement-based quantum computation on cluster states”. In: *Physical Review A* 68.2 (Aug. 2003). ISSN: 1094-1622. DOI: [10.1103/physreva.68.022312](https://doi.org/10.1103/physreva.68.022312). URL: <http://dx.doi.org/10.1103/PhysRevA.68.022312>.
- [Sho97] Peter W. Shor. “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer”. In: *SIAM Journal on Computing* 26.5 (Oct. 1997), pp. 1484–1509. ISSN: 1095-7111. DOI: [10.1137/s0097539795293172](https://doi.org/10.1137/s0097539795293172). URL: <http://dx.doi.org/10.1137/S0097539795293172>.
- [SK17] Adam Sawicki and Katarzyna Karnas. “Criteria for universality of quantum gates”. In: *Physical Review A* 95.6 (June 2017). ISSN: 2469-9934. DOI: [10.1103/physreva.95.062303](https://doi.org/10.1103/physreva.95.062303). URL: <http://dx.doi.org/10.1103/PhysRevA.95.062303>.
- [SSY25] Nadish de Silva, Wilfred Salmon, and Ming Yin. “Fast algorithms for classical specifications of stabiliser states and Clifford gates”. In: *Quantum* 9 (Jan. 2025), p. 1586. ISSN: 2521-327X. DOI: [10.22331/q-2025-01-08-1586](https://doi.org/10.22331/q-2025-01-08-1586). URL: <https://doi.org/10.22331/q-2025-01-08-1586>.
- [Van+06] Maarten Van den Nest et al. “Universal Resources for Measurement-Based Quantum Computation”. In: *Phys. Rev. Lett.* 97 (15 Oct. 2006), p. 150504. DOI: [10.1103/PhysRevLett.97.150504](https://link.aps.org/doi/10.1103/PhysRevLett.97.150504). URL: <https://link.aps.org/doi/10.1103/PhysRevLett.97.150504>.
- [VDD04] Maarten Van den Nest, Jeroen Dehaene, and Bart De Moor. “Graphical description of the action of local Clifford transformations on graph states”. In: *Physical Review A* 69.2 (Feb. 2004). ISSN: 1094-1622. DOI: [10.1103/physreva.69.022316](https://doi.org/10.1103/physreva.69.022316). URL: <http://dx.doi.org/10.1103/PhysRevA.69.022316>.

- [VDD05] Maarten Van den Nest, Jeroen Dehaene, and Bart De Moor. “Local unitary versus local Clifford equivalence of stabilizer states”. In: *Physical Review A* 71.6 (2005). ISSN: 1094-1622. DOI: [10 . 1103 / physreva . 71 . 062323](https://doi.org/10.1103/PhysRevA.71.062323). URL: [http : //dx.doi.org/10.1103/PhysRevA.71.062323](http://dx.doi.org/10.1103/PhysRevA.71.062323).
- [Wet20] John van de Wetering. *ZX-calculus for the working quantum computer scientist*. 2020. arXiv: [2012.13966 \[quant-ph\]](https://arxiv.org/abs/2012.13966). URL: <https://arxiv.org/abs/2012.13966>.

A Appendix: Background on Quantum Computing

A.1 Qubits

DEFINITION A.1 (QUBIT) A single **qubit** $|\phi\rangle$ is a state

$$|\phi\rangle = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} \in \mathbb{C}^2, \quad (\text{A.1})$$

where $|\alpha|^2 + |\beta|^2 = 1$ and $\alpha, \beta \in \mathbb{C}$. We say $|\phi\rangle$ is in a **superposition** between $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$. Thus a single qubit is a length 2 vector in the Hilbert space $\mathbb{C}^2$ with a ℓ_2 norm of 1.

DEFINITION A.2 (BRAKET NOTATION) Define “kets” $|0\rangle := \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $|1\rangle := \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, both in $\mathbb{C}^2$. For any $|\phi\rangle$, let “bra” $\langle\phi| := |\phi\rangle^\dagger = (|\phi\rangle^*)^T$ be its conjugate transpose. Thus the braket denotes the inner product:

$$\langle\phi|\phi\rangle := \langle\phi| \cdot |\phi\rangle. \quad (\text{A.2})$$

Now the ℓ_2 norm of $|\phi\rangle$ can now be written as $\| |\phi\rangle \|_2 = \sqrt{\langle\phi|\phi\rangle}$.

DEFINITION A.3 (TENSOR PRODUCT) Let $A \in \mathbb{C}^{m \times n}$ and $B \in \mathbb{C}^{o \times p}$. For our purposes, it suffices to define the **tensor product**

$$A \otimes B := \begin{bmatrix} a_{1,1}B & \cdots & a_{1,n}B \\ \vdots & \ddots & \vdots \\ a_{n,1}B & \cdots & a_{n,m}B \end{bmatrix} \quad (\text{A.3})$$

where $a_{i,j}$ denotes the i th row and j th column of A .

DEFINITION A.4 (MULTIPLE QUBITS) Let $x = \overline{x_1 x_2 \cdots x_d} \in \{0, 1\}^d$, i.e. a binary bit-string. Define

$$|x\rangle := \bigotimes_{i=1}^d |x_i\rangle. \quad (\text{A.4})$$

For example, $|00\rangle = |0\rangle \otimes |0\rangle = (1 \ 0 \ 0 \ 0)^T$. We may also replace x by its integer evaluation as a binary number, by abuse of notation. For example,

$$|3\rangle = |11\rangle = (0 \ 0 \ 0 \ 1)^T \quad (\text{A.5})$$

but we may also have

$$|3\rangle = |011\rangle = (0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0)^T \quad (\text{A.6})$$

Which binary representation we choose should be clear from context. For example, this gives us two equivalent definitions of a q -qubit state

$$|\varphi\rangle = \sum_{x \in \{0,1\}^d} \alpha_x |x\rangle = \sum_{x \in [2^d] \cup \{0\}} \alpha_x |x\rangle \in \mathbb{C}^{2^d} \quad (\text{A.7})$$

where $\alpha_x \in \mathbb{C}$ and $\sum_{x \in \{0,1\}^d} \alpha_x |x\rangle = 1$. As the number of dimensions increases exponentially with regards to the number of qubits, bracket notation will become increasingly concise compared to traditional vector notation.

DEFINITION A.5 (TENSOR PRODUCT NOTATION) For brevity, we will write $|\phi\rangle |\psi\rangle = |\phi\rangle \otimes |\psi\rangle$, so adjacent kets imply $\otimes$. However, note that $|\phi\psi\rangle \neq |\phi\rangle |\psi\rangle$ in general. Also, let

$$A^{\otimes n} := \bigotimes_{i=1}^n A. \quad (\text{A.8})$$

DEFINITION A.6 Vectors $|v_1\rangle, \dots, |v_n\rangle \in \mathbb{C}^n$ are an orthonormal basis if

1. $\|v_i\| = 1$ for every $i \in [n]$ and
2. $|v_i\rangle \perp |v_j\rangle$ (equivalently, $\langle v_i | v_j \rangle = 0$) for every $i, j \in [n]$ with $i \neq j$.

Henceforth, all bases are orthonormal unless explicitly said otherwise.

DEFINITION A.7 (MEASUREMENT) The **Born Rule** defines how quantum measurement occurs: An n -qubit state $|\phi\rangle$ can be measured under any orthonormal basis $\{|v_i\rangle\}_{i \in [2^n]}$ of $\mathbb{C}^{2^n}$. When measured, the output i occurs with probability $|\langle v_i | \phi \rangle|^2$. After seeing outcome i , the state collapses to $|v_i\rangle$. This process is irreversible.

DEFINITION A.8 (PARTIAL MEASUREMENT) One can also view measurement as projection. This is useful for **partial measurement**, where only some qubits in a state are measured. Let the n -qubit space $\mathbb{C}^{2^n} = S_1 \otimes S_2$, where $S_1 \cong (\mathbb{C}^2)^{\otimes k}$. Let $\{|s_i\rangle\}_{i \in [2^k]}$ be an orthonormal basis for S_1 . On outcome $i \in [2^k]$, define the **projection matrix** on S_1 as

$$P_i := |s_i\rangle \langle s_i|, \quad \text{such that} \quad P_i = P_i^\dagger, \quad P_i^2 = P_i. \quad (\text{A.9})$$

Define the **partial measurement** on S_1 by $M_i = P_i \otimes I \in \mathbb{C}^{2^n \times 2^n}$. For a state $|\phi\rangle \in \mathbb{C}^{2^n}$,

$$\Pr(\text{measuring } i \text{ from } |\phi\rangle) = \langle \phi | M_i | \phi \rangle = \|M_i |\phi\rangle\|^2. \quad (\text{A.10})$$

On measuring outcome i , the post-measurement state becomes

$$|\phi_i\rangle = \frac{M_i |\phi\rangle}{\sqrt{\langle \phi | M_i | \phi \rangle}}. \quad (\text{A.11})$$

EXAMPLE A.9 Consider the states

$$|+\rangle := \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle, \quad |-\rangle := \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle. \quad (\text{A.12})$$

Measuring either in the **standard basis** $\{|0\rangle, |1\rangle\}$ gives $|0\rangle$ with $\frac{1}{2}$ probability since

$$|\langle 0|+\rangle|^2 = \left| \frac{1}{\sqrt{2}} \cdot 1 + \frac{1}{\sqrt{2}} \cdot 0 \right|^2 = \frac{1}{2} \quad (\text{A.13})$$

and similarly for $|1\rangle$. The squaring makes $|-\rangle$ proceed similarly as well.

DEFINITION A.10 A two qubit state $|\phi\rangle$ is **entangled** if it cannot be written as the tensor product of single qubits. That is, $\nexists |\psi\rangle, |\varphi\rangle \in \mathbb{C}^2$ such that $|\phi\rangle = |\psi\rangle \otimes |\varphi\rangle$. If a state is not entangled, it is called **separable**. An n qubit state $|\psi\rangle$ is entangled (or *contains* entanglement) if it is not fully separable.

A.2 Quantum Gates

DEFINITION A.11 A 1-qubit quantum gate U is a matrix $U \in \mathbb{C}^{2 \times 2}$ with:

1. $U^\dagger U = U U^\dagger = I$;
2. equivalently, U is a unitary matrix;
3. equivalently, $U = |\phi_0\rangle\langle 0| + |\phi_1\rangle\langle 1|$, where $\{|\phi_0\rangle, |\phi_1\rangle\}$ form an orthonormal basis.

Common single qubit gates are given in Table 2.

THEOREM A.12 (EULER DECOMPOSITION) Every one-qubit unitary U can be written as

$$U = e^{i\alpha} R_z(\beta) R_x(\gamma) R_z(\delta), \quad (\text{A.14})$$

where $\alpha, \beta, \gamma, \delta \in [0, 2\pi)$.

Proof. This is a standard proof, seen for example in Section 4.2 of [MikeIke]. Since U is unitary, $|\det U| = 1$. Choose $\alpha \in [0, 2\pi)$ such that $\det U = e^{2i\alpha}$. Let $V := e^{-i\alpha} U$ satisfy $\det V = 1$ while also being unitary. Now let

$$V = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad (\text{A.15})$$

Since the columns of V are orthonormal, the second column must be a unit vector orthogonal to the first. Thus

$$\begin{pmatrix} b \\ d \end{pmatrix} = \lambda \begin{pmatrix} -\bar{c} \\ \bar{a} \end{pmatrix} \quad (\text{A.16})$$

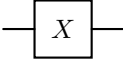
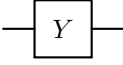
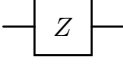
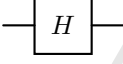
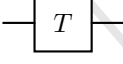
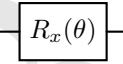
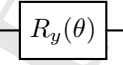
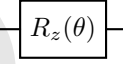
Gate	Circuit	Matrix
X		$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
Y		$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$
Z		$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$
H		$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
S		$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$
T		$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$
$R_x(\theta)$		$\begin{bmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$
$R_y(\theta)$		$\begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix}$
$R_z(\theta)$		$\begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$

Table 2: Common single qubit quantum gates

for some $|\lambda| = 1$. Hence

$$V = \begin{pmatrix} a & -\lambda\bar{c} \\ c & \lambda\bar{a} \end{pmatrix}. \quad (\text{A.17})$$

Taking determinants gives

$$1 = \det V = \lambda|a|^2 + \lambda|c|^2 \quad (\text{A.18})$$

$$= \lambda(|a|^2 + |c|^2) \quad (\text{A.19})$$

$$= \lambda. \quad (\text{A.20})$$

This also gives $|a|^2 + |c|^2 = 1$. Choose $\gamma, \eta, \mu \in \mathbb{R}$ such that

$$a = e^{i\eta} \cos(\gamma/2), \quad (\text{A.21})$$

$$c = e^{i\mu} \sin(\gamma/2). \quad (\text{A.22})$$

Then

$$V = \begin{pmatrix} e^{i\eta} \cos(\gamma/2) & -e^{-i\mu} \sin(\gamma/2) \\ e^{i\mu} \sin(\gamma/2) & e^{-i\eta} \cos(\gamma/2) \end{pmatrix}. \quad (\text{A.23})$$

On the other hand, direct multiplication gives

$$R_z(\beta)R_x(\gamma)R_z(\delta) = \begin{pmatrix} e^{-i(\beta+\delta)/2} \cos(\gamma/2) & -ie^{-i(\beta-\delta)/2} \sin(\gamma/2) \\ -ie^{i(\beta-\delta)/2} \sin(\gamma/2) & e^{i(\beta+\delta)/2} \cos(\gamma/2) \end{pmatrix}. \quad (\text{A.24})$$

Thus it is enough to choose β, δ so that

$$e^{i\eta} = e^{-i(\beta+\delta)/2}, \quad (\text{A.25})$$

$$e^{i\mu} = -ie^{i(\beta-\delta)/2}. \quad (\text{A.26})$$

Equivalently, modulo 2π , solving gives

$$\beta = \mu - \eta + \frac{\pi}{2}, \quad (\text{A.27})$$

$$\delta = -\mu - \eta - \frac{\pi}{2}. \quad (\text{A.28})$$

Therefore $V = R_z(\beta)R_x(\gamma)R_z(\delta)$. Since $U = e^{i\alpha}V$, we get

$$U = e^{i\alpha}R_z(\beta)R_x(\gamma)R_z(\delta) \quad (\text{A.29})$$

as desired. $\square$

FACT A.13 For matrices A, B, C, D over $\mathbb{C}$, scalar $\lambda \in \mathbb{C}$, and vectors $|\phi\rangle$ and $|\psi\rangle$ over $\mathbb{C}$, the tensor product satisfies:

1. Bi-linearity: $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$, so $(A \otimes B)(|\phi\rangle \otimes |\psi\rangle) = (A|\phi\rangle) \otimes (B|\psi\rangle)$.
2. Scalars commute: $\lambda(A \otimes B) = (\lambda A) \otimes B = A \otimes (\lambda B)$.

3. Associativity: $(A \otimes B) \otimes C = A \otimes (B \otimes C)$.

4. Unitary-preserving: If A, B are unitary matrices, $A \otimes B$ is unitary.

DEFINITION A.14 (MULTI-QUBIT GATES) An n -**qubit gate** is a unitary matrix in $\mathbb{C}^{2^n \times 2^n}$. If gates $U_1, \dots, U_k$ satisfy $U = \bigotimes_{i=1}^k U_i \in \mathbb{C}^{2^n}$, by Fact A.13.4 U is an n -qubit gate. Special named multi-qubit gates are given in Figure 3.

DEFINITION A.15 (CONTROLLED GATES) Many of the gates in Table 3 are **controlled** gates. Informally, controlled gates such as CNOT only apply their target operation when all controls $\text{---}\bullet\text{---}$ are in the state $|1\rangle$. Formally, for any n -qubit gate U , the singly controlled gate, denoted $C-U$, is defined by

$$C-U := |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes U. \quad (\text{A.30})$$

This definition generalizes naturally to multiple controls.

REMARK A.16 By the above definition, CNOT is just $C-X$, and Toffoli is $CC-X$. Note that we may suppress the hyphen when the meaning is clear.

The gate $C-Z$ is special because either qubit can be interpreted as the control: it only applies a phase when both qubits are in the state $|1\rangle$. To see this, note that

$$I = |0\rangle\langle 0| + |1\rangle\langle 1|, \quad Z = I - 2|1\rangle\langle 1|. \quad (\text{A.31})$$

Therefore,

$$C-Z = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes (I - 2|1\rangle\langle 1|) \quad (\text{A.32})$$

$$= I \otimes I - 2|11\rangle\langle 11|. \quad (\text{A.33})$$

Thus $C-Z$ acts as the identity on every computational basis state except $|11\rangle$, where it applies a phase of -1 .

Gate	Circuit	Matrix
CNOT		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$
CCNOT or Toffoli		$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$
SWAP		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
C-Z		$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

Table 3: Common multi-qubit quantum gates

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